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Rock Piles as Kōura Habitat

Habitat enhancement for a native freshwater crayfish threatened by an invasive predator: a field trial of stone pile structures in an Aotearoa New Zealand lake

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Abstract

1. Kōura (*Paranephrops planifrons*), Aotearoa New Zealand’s endemic freshwater crayfish, have declined by up to 96 % in Lake Rotoiti (Te Arawa Lakes, Rotorua), driven by predation from the invasive brown bullhead catfish and by thermal stratification and invasive macrophytes that compress suitable habitat into the shallow littoral zone. Because these stressors cannot be removed at the lake scale, adding structural refuge may be one of few practical management options.
2. We tested whether stone pile structures deployed in the littoral zone of Lake Rotoiti could provide structural refuge for kōura from predation. Kōura and catfish were monitored at five stone piles, five control, and five paired natural rocky reference sites for 18 months using a quasi before-after control-impact (quasi-BACI) design, with kōura occupancy modelled using a Bayesian framework that incorporated detection probabilities from baited traps and fyke nets.
3. Stone pile construction produced no statistically significant change in kōura abundance, biomass, or occupancy relative to controls, but effects consistently trended positive across models; a simulation-based power analysis showed the study was underpowered to detect the observed effect size, given high pre-construction occupancy (81–89%) at all site types and only five site pairs.
4. Kōura use of stone piles was strongly seasonal: catch rates at stone pile and control sites fell in winter as kōura withdrew to deeper water, while reference sites remained stable. Catfish showed no comparable attraction to stone piles in any season or model.
5. Synthesis and applications. Stone piles show early, consistent evidence of benefiting a threatened native crayfish without benefiting its invasive predator, and are most valuable when habitat compression and predation pressure peak together in summer and autumn — conditions expected to intensify under climate warming. This study establishes a tested, scalable construction and consenting methodology that lake managers can deploy as an interim refuge-based tool while catfish eradication remains unachievable at the lake scale; we recommend continued monitoring beyond three years and trialling connected, corridor-style structures rather than isolated piles.

Introduction

Freshwater biodiversity is declining faster than any other biome, with one quarter of freshwater fauna now threatened with extinction (Sayer et al., 2025). Among the most endangered are

1 freshwater crayfish, with 32% of species assessed as threatened globally due to habitat loss,
2 invasive species, and declining water quality (Richman et al., 2015). As keystone species and
3 ecosystem engineers, crayfish play crucial roles in nutrient cycling, bioturbation, and food web
4 dynamics, and their loss can have cascading consequences for freshwater ecosystems (Collier
5 et al., 1997; Momot, 1995; Parkyn et al., 2001). In Aotearoa New Zealand, these pressures
6 converge on kōura (*Paranephrops planifrons* White, 1842), an endemic freshwater crayfish now
7 facing rapid population decline in lakes that have long supported some of the largest customary
8 kōura fisheries in the country (Hiroa, 1921; Kusabs & Quinn, 2009).

9 Kōura populations in Lake Rotoiti, located in the Rotorua Te Arawa Lakes region of the North
10 Island of Aotearoa New Zealand (hereafter Aotearoa), have declined by up to 96 % over the past
11 two decades (I. A. K. Kusabs et al., 2026). This collapse has virtually ended customary harvest
12 practices for Te Arawa iwi (Māori tribal group) who are hunga tiaki (traditional custodians)
13 of the lake (Kusabs, Hicks, et al., 2015). The decline in kōura numbers is driven by multiple
14 interacting stressors that compound over time in different parts of the lake (I. A. K. Kusabs et
15 al., 2026; Lee et al., 2025). The invasive brown bullhead catfish (*Ameiurus nebulosus* Lesueur,
16 1819, hereafter catfish), confirmed in Lake Rotoiti in 2016 (Francis, 2019; Grayling, 2020),
17 preys directly on kōura and suppresses their foraging through non-consumptive ‘fear’ effects
18 (Raven et al., 2026a). Predation pressure from catfish is compounded by additional stressors
19 that constrain where in the lake kōura can seek refuge from this invasive predator (I. A. K.
20 Kusabs et al., 2026).

21 During the day, kōura shelter in complex rocky habitat and in deeper water to avoid predation
22 (Jowett et al., 2008; Kusabs, Quinn, et al., 2015), moving out of both at night to forage in
23 the productive littoral zone (Devcich, 1979). The importance of deep-water refuge is shown in
24 Lake Taupō, where kōura persist at a depth even in the presence of catfish because the lake
25 remains oxygenated below the thermocline (I. Kusabs & Taiaroa, 2015; Verburg & Albert,
26 2019). In Lake Rotoiti, this deep refuge becomes anoxic during summer stratification and
27 unavailable for kōura, compressing the zone of liveable habitat into the shallow littoral margins
28 (I. A. K. Kusabs et al., 2026), precisely where catfish are most active (Blanchfield et al., 2026;
29 Dedual, 2002). Dense beds of invasive macrophytes further restrict kōura movement through
30 this compressed zone and intensify oxygen depletion during die-off events (Hamilton et al.,
31 2005; Kusabs & Quinn, 2009; Vincent et al., 1984). Catfish pressure and thermal stratification
32 are expected to only strengthen with predicted climate scenarios (Lee et al., 2025; Woolway
33 et al., 2021), making kōura habitat compression and predation pressure progressively more
34 severe.

1 The compression of suitable habitat and subsequent concentration of kōura into the littoral
2 zone makes the quality of nearshore lacustrine habitats increasingly important for remaining
3 populations. In stream environments, kōura are positively associated with structurally complex
4 habitats including bank undercuts, woody debris, leaf litter, and coarse substrates (Jowett
5 et al., 2008; Parkyn et al., 2009; Parkyn & Collier, 2004; Usio & Townsend, 2000). In lake
6 environments, benthic substrate characteristics are a stronger predictor of kōura distribution
7 than nutrient enrichment or predator abundance (Kusabs, Quinn, et al., 2015), and kōura
8 are positively associated with coarse substrates and riparian vegetation (Raven et al., 2026b).
9 Coarse rocky substrate provides refugia from predators and shelter during moulting (Streissl
10 & Hödl, 2002). Experimental studies confirm that shelter availability directly reduces catfish
11 predation on crayfish (Adams, 2007; Zhao et al., 2024), and kōura increase refuge use in
12 the presence of catfish compared to native predators (Raven et al., 2026a), emphasising the
13 importance of shelter availability for kōura populations. Shelter availability is likely a limiting
14 factor for kōura in Lake Rotoiti, where the majority of the shoreline is sandy with dense beds
15 of introduced macrophytes and little rocky structure. Adding rocky substrate to soft-bottomed
16 lake areas has been shown to significantly increase crayfish abundance (Johnsen & Taugbøl,
17 2008), and increasing structural complexity in streams improves crayfish restocking success
18 (Huolila et al., 1997). Deploying habitat enhancement structures that add structural complexity
19 in the littoral zone could potentially be a practical way to restore kōura populations in Lake
20 Rotoiti.

21 In this study habitat enhancement structures in the shape of stone piles were deployed along
22 the shoreline of Lake Rotoiti to investigate the ability of the structures in providing additional
23 habitat for kōura to live and hide from predators. Kōura presence, abundance, and biomass
24 were monitored at stone pile, control, and reference sites over 18 months. We predicted that
25 kōura occupancy, abundance and biomass would increase more rapidly at the stone pile sites
26 than at control sites, as intense catfish predation pressure in the littoral zone was expected
27 to compel kōura actively toward available refuge structures. We further predicted that kōura
28 abundance would continue to increase progressively after construction, at least within the first
29 18 months, and that catfish would instead show seasonal patterns of shoreline use rather than
30 a construction-driven trend (Barnes, 1996; Blanchfield et al., 2026; Dedual, 2002). Finally, we
31 predicted that stone pile sites would progressively converge toward kōura densities observed at
1 natural rocky reference sites, and that catfish would not exhibit a similar pattern.

2 Materials and Methods

3 Study location and construction

4 Lake Rotoiti (-38.03, 176.42) is a warm monomictic mesotrophic lake located in the Rotorua
5 Te Arawa Lakes region of the North Island of Aotearoa (34 km², maximum depth 128 m).
6 The lake stratifies for approximately nine months a year (Figure 7), with pronounced thermal
7 stratification compressing oxygenated habitat into the shallow littoral zone during summer
8 and autumn. Since 2008, nutrient-rich inflows from Lake Rotorua have been diverted toward
9 the Kaituna River via the Ōhau Channel diversion wall to improve water quality in Lake
10 Rotoiti (Hamilton et al., 2005; Prentice et al., 2026). Lake level is regulated by a weir at the
11 outlet of the Kaituna River, providing a stable and controlled water level throughout the year.
12 Catfish are present throughout the lake but occur in particularly high concentrations in Te
13 Weta Bay on the Northwest side of the lake (Francis, 2019). Five stone pile sites were selected
14 alongside the lake shoreline based on accessibility and landowner approval (Figure 1). At each
15 stone pile site, a paired control site was established 50 m along the shoreline, matched for
16 habitat characteristics e.g., water depth, substrate type and vegetation cover. Five naturally
17 occurring rocky shoreline locations, with each two monitoring sites 50 m apart were selected
18 as reference. In total, 20 sites of each 150 m² (15 × 10 m) were monitored seasonally over 18
19 months. Physical characteristics of all sites are summarised in Table 1. Baseline monitoring was
20 conducted in February 2025 prior to stone pile deployment, which occurred the following week.
21 Stone piles were constructed from locally sourced quarry rock (Manawahe Stone and Quarry,
22 ~25 km from the lake) with a diameter ranging from 85 - 315 mm (mean = 173 mm). At each
23 of the five stone pile sites, three stone piles were arranged horizontally along the shoreline by
24 hand from the shore. Stone pile design was informed by mesocosm experiments demonstrating
25 kōura preference for enclosed stone structures (unpublished data). Stone piles were positioned
26 approximately 4.3 m from shore at a mean depth of 0.49 m and with a mean area of 4.1 m².

27 Monitoring

1 Physical, chemical, and biological parameters were recorded at each site during each monitoring
2 round. Static site characteristics including substrate composition, slope to depth contour,
3 riparian vegetation, overhanging tree cover, and wood cover were visually estimated during
4 baseline monitoring only, as these were not expected to change substantially over the monitoring
5 period. Substrate composition was visually estimated by a single observer walking around

each site and using an underwater viewing chamber, categorising bedrock, boulders (>256 mm), cobble (64–256 mm), gravel (2–64 mm), sand (0.063–2 mm), mud (<0.063 mm), and organic matter (Wentworth, 1922). A substrate index, that indicates suitability for koura, was calculated from these estimates (Jowett et al., 2008; Jowett & Richardson, 1990):

$$\begin{aligned} \text{Substrate index} = & 0.08 \times \text{bedrock} + 0.07 \times \text{boulders} \\ & + 0.06 \times \text{cobble} + 0.04 \times \text{gravel} + 0.03 \times \text{sand} \\ & + 0.02 \times \text{organic matter} + 0.01 \times \text{mud} \end{aligned}$$

Coarser substrates were assigned higher weights as they provide greater interstitial pore space and stable structures used by koura (Nyström et al., 2006; Usio & Townsend, 2000).

After stone pile construction, measurements of the size and dimensions of each structure were taken. The post-construction substrate index for stone pile sites was recalculated using each site’s combined stone-pile footprint area, expressed as a percentage of the 150 m² site area and treated as 80 % cobble and 20 % boulders displacing an equivalent area of sand, to reflect their contribution to substrate complexity. Slope was assessed as the horizontal distance from the shoreline to the 20 m depth contour, derived from bathymetric maps and measured in a projected coordinate system (NZTM2000; EPSG:2193).

Seasonal variable parameters were recorded at every monitoring round and included aquatic vegetation cover (emergent and submerged macrophytes). Chemical parameters included water temperature (°C), dissolved oxygen (mg L⁻¹), and conductivity (μS cm⁻¹) were measured using a YSI ProSolo meter (Pro2030, YSI Inc., Yellow Springs, OH, USA). pH was measured using a handheld pH meter (EC-PCTestr35, Eutech Instruments).

Koura and fish were sampled every monitoring round using two single-winged, double-throated fyke nets (600 mm high D-shaped entrance hoop) and two collapsible bait traps (480 × 250 mm; two 50 mm entrance openings) baited with catfish flesh (Joy et al., 2013; MacNeil et al., 2025). Fyke nets were placed perpendicular to the shoreline outside of the stone pile structures or at 10 m intervals, with bait traps positioned between them. All equipment was deployed simultaneously at each site for a 24-h period. Captured koura were measured for orbital carapace length (OCL) using a digital calliper (0.01 mm precision; 150 mm) and for wet mass using a digital scale (± 1 g; 5 kg capacity, Anko Brand, Kmart Group, Mulgrave, VIC, Australia). Catfish and other fish species, were recorded for abundance, measured for total length, and weighed for wet mass. Catch per unit effort (CPUE) and biomass per unit effort (BPUE) were calculated for all koura and fish based on the number of nets and traps

5 deployed at each site per monitoring round, standardised by soak time (Kusabs, Quinn, et
6 al., 2015). Kōura weights were unavailable for 36 individuals because of scale failure and were
7 therefore estimated from a log $-\log$ length–weight regression fitted to individuals with both
8 measurements ($n = 840$), following standard approaches for weight length relationships in
9 aquatic organisms Figure 8 (Froese, 2006). Predicted values were back-transformed with a
10 lognormal bias correction factor $10^{0.5\sigma^2}$ (Quinn & Keough, 2002).

$$\log_{10}(\text{Weight [g]}) = -2.67 + 2.684 \times \log_{10}(\text{OCL length [mm]})$$

11 Data analysis

12 Data analysis was a three-part process, the first assessed baseline equivalence between stone
13 pile, control, and reference sites before construction, the second assessed the impact of stone
14 pile construction and after-construction trajectory on kōura and catfish occupancy, abundance,
15 and biomass, and the third assessed whether stone pile sites converged toward reference sites
16 and if reference sites remained stable over time. All analyses were conducted in R (version
17 4.6.1; R Core Team (2026)) using the glmmTMB and emmeans packages (Brooks et al., 2017;
18 Lenth & Piaskowski, 2017).

19 Covariate selection for all models followed a consistent procedure. Corrected Akaike Information
20 Criterion (AICc) values for each base model were compared against models with additional
21 covariates (temperature, season, substrate index, maximum effective fetch, or wave energy
22 proxy) added at a time, and a covariate was retained only when it reduced AICc by more than
23 two units relative to base model (Burnham & Anderson, 2002). Robustness and sensitivity for
24 all models were assessed by evaluating uniform distribution, dispersion and outlier rate using
25 the DHARMA package (Hartig, 2016). Across the full set of models, the Benjamini–Hochberg
26 correction method was applied to control the false discovery rate within each of the three
27 analysis components.

28 Site variation before stone pile construction was assessed using paired Wilcoxon signed-rank
29 tests to compare stone pile sites with their paired control sites, and unpaired Wilcoxon rank-sum
30 tests to compare stone pile sites with reference sites. This was done for response variables
31 (water quality, substrate, and kōura and catfish CPUE and BPUE) to confirm that the stone
32 pile and control sites were equivalent before construction, and to characterise the difference
33 between stone pile and reference sites. To further assess baseline equivalence, generalised

1 linear models (GLMs) were fitted to test whether kōura and catfish CPUE and BPUE differed
2 between site type (stone pile / control) and between the two reference sites (reference 1 /
3 reference 2) using a Tweedie family with a log link. Following the covariate selection procedure,
4 temperature was included in the catfish models comparing reference sites, as the base model
5 failed to converge and temperature was required for numerical stability.

6 The second component of the analysis tested construction impact relative to control sites using
7 a quasi-before-after, control-impact (quasi-BACI) design ([Underwood, 1994](#)), and separately
8 tested after-construction trajectory of kōura and catfish populations. The quasi-BACI design
9 compared one pre-construction monitoring round with six post-construction monitoring rounds
10 using generalised linear mixed effects models (GLMMs) to test the effect of stone pile construc-
11 tion on kōura and catfish occupancy, abundance, and biomass. Time period (before / after)
12 and site type (stone pile / control) were included as an interaction term, monitoring round
13 as a covariate and site nested within control-impact pair as a random effect. Occupancy was
14 modelled using a binomial GLMM, and abundance and biomass using a Tweedie GLMM with a
15 log link; fetch was retained as a covariate in the catfish model. The post-construction trajectory
16 models used the same response variables, families, covariate, and random effect structure, but
17 with site type and monitoring round included as an interaction term in place of the before /
18 after comparison.

19 The third component of the analysis assessed whether stone pile sites converged toward reference
20 sites over time, and whether reference sites themselves remained stable. Kōura and catfish
21 abundance and biomass were tested using GLMMs with site type (stone pile / reference) and
22 monitoring round included as an interaction term, and site number as a random effect, using
23 a Tweedie family with log link; temperature was retained as a covariate in the catfish model.
24 Reference site stability was tested in GLMMs with site type (reference 1 / reference 2) and
25 monitoring round as an interaction term, and site nested within pair as a random effect, again
26 using Tweedie family with log link; season was retained as a covariate in the kōura model and
27 temperature was retained as a covariate in the catfish model.

28 To complement the GLMM abundance analysis, kōura site-level occupancy was estimated using
29 a two-method occupancy model fitted within a Bayesian framework. Because bait traps and
30 fyke nets provide independent and simultaneous detection opportunities at each site, true site
31 occupancy could be separated from imperfect and method-specific detection probabilities. A
32 custom Stan observation model was implemented using the brms package ([Bürkner, 2017, 2018](#))
33 using cmdstanr ([Gabry et al., 2025](#)) to estimate true occupancy and detection probabilities
34 for baited traps and fyke nets via logit links. Standardised water temperature was included

1 as the covariate for detection probability. Three occupancy sub-models were fitted. A full
2 model across all monitoring rounds served as baseline comparison. A summer-only model and
3 a warm-season model (summer and autumn combined) were fitted to focus on the period of
4 highest thermal stratification, when kōura are known to disproportionately occupy nearshore
5 littoral habitats (I. A. K. Kusabs et al., 2026) and when catfish predation pressure is highest
6 in the areas where the stone piles are located (Blanchfield et al., 2026). The warm-season
7 model maximises ecological relevance, while retaining a tractable formula with season as an
8 additive covariate. All models were run with four chains of 4000 iterations (1500 warmup) and
9 convergence was assessed using R values (threshold < 1.01) and bulk effective sample sizes
10 (threshold > 400) (Vehtari et al., 2021).

11 A simulation-based power analysis was conducted to quantify the sample size required for
12 statistical detection of the BACI effect under the observed effect size and variance structure,
13 and to put the non-significant results in perspective relative to study design rather than true
14 absence of effect. For each analysis the occupancy and abundance, datasets were simulated
15 across a grid of scenarios varying in number of site pairs (3-30) and after-period monitoring
16 rounds (2-12). For the occupancy BACI, simulated binary detection data were generated
17 using values from the warm-season occupancy model posterior means, and each simulated
18 dataset was analysed using a logistic GLMM (lme4; Bates et al. (2015)). For the CPUE BACI,
19 simulated abundance data were generated using fixed effects and variance components from the
20 kōura abundance BACI model, using a log-normal approximation to the Tweedie distribution,
21 and each dataset was analysed using a linear mixed model fitted to a log-transformed CPUE.
22 Statistical power was estimated as the proportion of simulations (300 per scenario) in which
23 the model's 95 % confidence interval for the BACI coefficient excluded zero in the positive
24 direction.

25 Results

26 Kōura presence was detected at stone pile sites within three months of construction, with
1 occupancy and abundance trending consistently higher at stone pile sites than paired control
2 sites throughout the 18-month post-construction period, though this difference did not reach
3 statistical significance. Results across all models and response variables are summarised in
4 Figure 2 and ?@tbl-models-summary-full-supp.

5 Baseline equivalence

6 Pre-construction, kōura abundance and biomass did not differ significantly between stone pile
7 and paired control sites (**?@tbl-baseline-combined-supp**), confirming baseline equivalences
8 ($p = 0.783$ and $p = 0.571$). Kōura abundance ($p = 0.968$) and biomass ($p = 0.968$) did not
9 differ between reference sites. Catfish abundance ($p = 0.004$) and biomass ($p = 0.002$) likewise
10 did not differ between stone pile and control sites pre-construction. However, catfish abundance
11 and biomass were significantly higher at Reference 1 than Reference 2 ($p = 0.004$ and $p =$
12 0.002 , respectively), despite both sites having comparable rocky substrate. This suggests catfish
13 distribution varies considerably between nearby sites with similar habitat characteristics.

14 Construction impact (BACI) and post-construction trajectory

15 Stone pile construction produced no detectable step-change in kōura abundance, biomass,
16 or occupancy relative to paired control sites over the 18-month monitoring period following
17 deployment (all $p_{\text{adj}} = 0.858$) (Figure 3). A matched-season comparison, comparing Summer
18 2025 (before construction) with Summer 2026 (one year after construction), produced the
19 same conclusion (kōura abundance: $p = 0.582$). Post-construction trajectories did not diverge
20 significantly between stone pile and control sites (all $p_{\text{adj}} = 0.215$). Kōura abundance declined
21 significantly across post-construction rounds at control and stone pile sites ($\beta = -0.41$, $p =$
22 0.004), with the sharpest decline observed in Winter 2026 (Figure 4). Stone pile construction
23 produced no detectable step-change or trajectory divergence in catfish abundance, biomass, or
24 occupancy relative to control sites across any model or time period (Figure 2).

25 Kōura occupancy was high across all site types before construction, 81–89 % in summer
26 (**?@tbl-psi-warm-supp**). After construction, occupancy increased marginally at all sites,
27 89–94 % in summer, thereby limiting our ability to detect a treatment effect. The BACI
28 contrast to compare how much stone pile sites changed relative to control sites was slightly
29 positive, but the credible intervals spanned zero with wide uncertainty (Figure 5), indicating no
30 detectable additional increase at stone pile sites. However, this result likely reflects a measured
1 ceiling rather than biological response. After-treatment contrast showed no significant difference
2 between site types, but the direction of the effect was consistent with early habitat use: stone
3 pile sites showed a positive contrast relative to control sites, while control sites showed a
4 negative contrast relative to reference sites, and stone pile sites were near zero relative to
5 reference sites (Figure 5).

6 The non-significant BACI results should be interpreted in the context of statistical power. For
7 occupancy, power was near zero across all simulated scenarios (maximum 8%), reflecting the
8 ceiling effect leaving no statistical room for a BACI signal (Figure 9). Post-construction power
9 analysis showed that achieving 80 % power with the current five site pairs would require an
10 effect size of approximately 1.8 log-units, almost 6 times the current observed effect (Figure 6).
11 If the estimated positive trajectory of +0.24 log-units per seasonal round continues, the effect
12 size may reach this detection threshold within approximately 6 additional monitoring rounds
13 (~1.5 years).

14 **Convergence toward reference sites and reference stability**

15 Stone pile sites maintained a consistently lower kōura abundance than reference sites throughout
16 the post-construction period, with no significant change in this gap over time ($\beta = -0.09$, $p =$
17 0.429 ; Figure 10). Kōura abundance declined across monitoring rounds at both stone pile and
18 reference sites ($\beta = -0.09$, $p = 0.107$). The Winter 2026 monitoring round showed a pronounced
19 decline at stone pile sites (CPUE 0.7) while reference sites maintained substantially higher
20 abundance (CPUE 1.7; Figure 4), a within-round contrast not captured by the linear trajectory
21 model. Water temperature at Winter 2026 was approximately 1.5°C lower than Winter 2025 but
22 consistent across site types, ruling out temperature as a site-specific driver. Visual inspection
23 of stone pile surfaces during Winter 2026 monitoring revealed substantially greater filamentous
24 algal accumulation compared to earlier rounds (?@fig-stone-piles). Catfish abundance and
25 biomass showed no significant convergence or divergence between stone pile and reference sites
26 over time (Figure 2).

27 **Discussion**

28 Stone pile structures were deployed in the littoral zone of Lake Rotoiti to test whether physical
29 refuge could reduce predation pressure on kōura by invasive catfish. Thermal stratification and
1 invasive macrophytes already compress suitable kōura habitat into the shallow littoral zone, and
2 with catfish eradication probably unachievable at the lake scale, habitat-based interventions
3 represent one of the few management options currently available to support this declining
4 endemic crayfish.

5 Short-term construction effect

6 Kōura catch rates were highly variable across all site types throughout the monitoring period,
7 with wide confidence intervals, likely reflecting genuine biological variability in kōura distribution.
8 This variability may reflect seasonal activity patterns, detection probability changes across
9 monitoring rounds, and natural heterogeneity in littoral zone use, rather than stable site
10 associations. Despite this variability, the Bayesian occupancy model revealed a consistent
11 positive directional trend, with stone pile sites showing higher occupancy than control sites
12 after construction, and occupancy at stone pile sites was comparable to that at natural rocky
13 reference sites across both summer and autumn post-deployment comparisons. The BACI
14 posterior contrast was positive, indicating that occupancy increased more at stone pile sites
15 than control sites following construction, consistent with the intended purpose of the stone
16 piles as refuge, though whether this reflects active refuge-seeking driven by catfish predation
17 pressure or gradual colonisation and population expansion into newly available habitats cannot
18 be determined from the current data. However, no statistically significant effect was detected
19 indicating that while the predicted directional response occurred, it did not reach detectable
20 magnitude within the 18-month monitoring period.

21 The absence of a significant effect is most likely explained by a combination of monitoring
22 duration, statistical power, and system dynamics rather than ineffectiveness of the structures.
23 Habitat establishment for freshwater crayfish can be inherently slow. Johnsen & Taugbøl (2008)
24 reported that crayfish abundance at a stone addition site took 13 years to reach reference
25 levels, and monitoring periods shorter than two years are often insufficient to detect significant
26 responses to habitat modification (Pilotto et al., 2018; Ruhí et al., 2012; Testa et al., 2011).
27 Following reconstruction of the Lake Rotorua lakefront, which is located in close proximity
28 to our study site, kōura showed no meaningful recolonisation after 28 months (Kusabs, 2023).
29 They attributed slow recolonisation to persistent catfish predation pressure and distance from
30 source population, consistent with time scales observed here. The power analysis indicates that
31 five site pairs are insufficient to reliably detect effects of the magnitude observed here within
32 18 months, establishing a minimum design threshold: future stone pile monitoring programmes
1 should plan for at least three years of monitoring and expect potential of high pre-construction
2 occupancy, leaving little statistical room for a measurable treatment effect. The simultaneous
3 elevation of kōura CPUE across all site types in Autumn 2025 further complicated detection of
4 an early treatment effect in the BACI contrast.

5 Beyond study design, features of the system itself may also explain the weak response. Kōura

6 abundance is highest in lakes without catfish ([Raven et al., 2026b](#)), suggesting predation
7 pressure may suppress kōura populations well below the carrying capacity of existing habitat.
8 If so, structural habitat availability may not currently be the primary limiting factor, and a
9 weak or delayed response to habitat addition would be expected. Additionally, if kōura are
10 already at carrying capacity of existing rocky habitat, stone piles add new capacity that will
11 only be filled gradually as the population expands, which is unlikely to be detectable within 18
12 months. In either case, a delayed rather than absent population response would be expected.
13 Isolated stone pile structures may also be difficult for kōura to locate, as distance between
14 isolated habitat patches is a key determinant of colonisation success ([Hanski, 1998](#)).

15 **Winter 2026**

16 Kōura CPUE at stone pile and control sites fell sharply in Winter 2026, returning to levels
17 close to the pre-deployment baseline from Summer 2025, while reference sites maintained
18 substantially higher abundance. This result at the control and stone pile sites is consistent with
19 known seasonal migration behaviour across lake depth profiles and kōura activity rates during
20 cooler water temperatures ([Devcich, 1979](#)). When lake thermal stratification breaks down in
21 winter, the hypolimnion re-oxygenates and deep-water habitat becomes accessible, allowing
22 kōura to redistribute along the depth gradient ([Devcich, 1979](#)). Kōura activity declines at low
23 water temperatures ([Devcich, 1979](#)), reducing the probability of trap entry independently of
24 true abundance. Water temperature in Winter 2026 was 1.5 °C lower than Winter 2025, though
25 this difference was consistent across all site types and therefore cannot explain why reference
26 sites were substantially more resilient to the winter decline than the stone pile sites.

27 Natural rocky reference sites in this study represent the ecological target for stone pile deploy-
28 ment. Reference sites had established kōura populations, connectivity to deeper waters, and
29 extensive refuge habitat, allowing kōura to maintain littoral presence through winter or return
30 rapidly from deeper refuges as conditions fluctuated. Stone pile sites are isolated structures with
31 less connectivity to deep and extensive refuge habitat, meaning kōura that withdrew in winter
32 had no quick pathway back to the structures. This difference likely explains why stone pile sites
33 maintained consistently lower kōura abundance than reference sites throughout the monitoring
1 period, with no significant convergence toward reference levels. However, kōura occupancy at
2 stone pile sites was comparable to that at reference sites after construction, suggesting that
3 kōura found and used the structures at similar frequency to natural rocky habitat but at lower
4 densities. This distinction between occupancy and abundance likely reflects the early stage of

5 colonisation and the small structural footprint of stone piles relative to the extensive natural
6 rocky habitat at reference sites. These differences are expected to narrow as biofilm succession
7 matures and kōura accumulate at the structures over time.

8 Extensive growth of filamentous algae was observed on the stone pile structures in Winter 2026
9 (?@fig-stone-piles) but did not occur to the same degree in Winter 2025. Filamentous algal
10 growth on hard substrates is characteristic of winter conditions in thermally stratified lakes,
11 as seasonal mixing increases nutrient availability in the water column (Wetzel, 2001), while
12 reduced water temperature suppresses invertebrate grazing activity (Bonacina et al., 2023),
13 allowing algal mats to accumulate on available hard surfaces. The stone pile surfaces, freshly
14 quarried and free from biofilm at deployment, provided clean substrate for early colonisers,
15 and by Winter 2026 a diverse assemblage had established, including cyanobacteria, green algae,
16 *Sinelobus stanfordi* (Richardson, 1901), Tardigrada, Oribatida, Chironomidae, Rhabdocoela,
17 Oligochaeta, and ciliates (*Paramecium* spp. and *Vorticella* spp.). Rather than representing
18 degradation of habitat quality, this succession is a natural trajectory, as temperatures rise in
19 spring and summer, grazer activity increases, stratification reduces nutrient loading (Bonacina
20 et al., 2023; Wetzel, 2001), and algal mat cover is expected to decline, making the stone piles
21 more accessible for kōura refuge.

22 Despite the near-zero trap catches in Winter 2026, night spotlighting at two stone pile sites in
23 August 2026 recorded one large kōura inside the structure at site 3 and three small individuals
24 in and around the structure at site 5, confirming that kōura had not abandoned the stone
25 piles. Non-detection by nets in winter may therefore reflect kōura sheltering within internal
26 crevices rather than true absence, as nets require active kōura movement to be effective. Taken
27 together, the Winter 2026 data suggest that stone pile use by kōura is inherently seasonal, as
28 these structures are most valuable during summer and autumn when thermal stratification
1 compresses suitable habitat into the littoral zone and catfish predation pressure peaks, while in
2 winter kōura redistribute to deeper water regardless of structure type. This seasonality defines
3 when stone pile structures are most needed.

4 Catfish

5 Catfish showed their seasonal patterns in littoral zone use, but these were unrelated to stone pile
6 presence. Stone pile construction produced no evidence of catfish attraction or accumulation.
7 Even if catfish were to use stone piles to some degree, the ecological significance of this would
8 likely be limited given the extensive availability of existing suitable habitat for catfish in Lake

9 Rotoiti, particularly invasive macrophyte beds (Barnes & Hicks, 2003). The single-site increase
10 in catfish CPUE during Summer 2026 was mirrored by an increase at reference sites and
11 was consistent with lake-wide seasonal redistribution into warm littoral waters rather than
12 any stone-pile-specific event (Blanchfield et al., 2026). Overall catfish catches remained low
13 throughout the study relative to targeted removal netting in Te Weta Bay, where highest catfish
14 densities are concentrated (Francis, 2019). In autumn, catfish shift to mid-water movement
15 across the lake basin remaining above 15 m depth (Blanchfield et al., 2026), reducing predation
16 pressure on kōura in the littoral zone.

17 **Implications for restoration design and monitoring**

18 Predator-prey interactions are shaped by habitat, which offers refuge to prey, influences how
19 easily predators can find and catch prey, and changes the costs and benefits of hunting for the
20 predator (Lennox et al., 2025). This framework is directly relevant to catfish-invaded lakes,
21 where habitat addition offers a way to influence this interaction. Rocky environments have been
22 identified as natural multi-threat refugia, capable of simultaneously buffering predation, thermal
23 extremes, and invasive species pressure across a wide range of taxa and ecosystems (Selwood &
24 Zimmer, 2020). In Lake Rotoiti, the two primary stressors driving kōura decline cannot be
25 managed at lake scale. Catfish eradication from a lake of this size is probably not achievable with
26 currently available methods, and targeted removal in high-density areas reduces local pressure
27 but cannot eliminate the lake-wide population (Dedual, 2019). Emerging biotechnologies such
28 as gene drives and host-specific biological control agents may offer future options (Kopf et
29 al., 2019; Teem et al., 2020; Thresher et al., 2014), but these remain experimental. Thermal
30 stratification will intensify as the climate continues to warm (Woolway et al., 2021), further
31 compressing kōura and expanding suitable conditions for catfish (Lee et al., 2025). When
32 the primary stressors cannot be removed, habitat addition becomes one of the few available
33 management tools, and refugia become most valuable under these conditions (Selwood &
1 Zimmer, 2020). While freshwater management increasingly recognises that multiple stressors
2 interact simultaneously (Orr et al., 2024; Spears et al., 2021), practical responses still tend
3 to focus on removing individual stressors. This study therefore tested whether structural
4 habitat addition can buffer the combined effects of predation and thermal compression without
5 removing either stressor.

6 The stone piles in this study were placed along the shoreline as isolated structures, separated
7 from each other and from deeper water by soft sandy substrate. Isolated habitat patches

8 support fewer species and are more vulnerable to local extinction than connected habitats
9 ([Hanski, 1998](#)). Placing stone pile structures perpendicular to the shoreline, running through
10 macrophyte beds and into deeper water, could create corridors connecting the shallow littoral
11 zone with deeper refugia and potentially allow kōura to move more freely between depth ranges
12 as conditions change just like on reference sites. This design has not been tested but represents
13 a promising direction for future deployments and could additionally suppress macrophyte
14 growth along the corridor.

15 The dense filamentous algal growth observed on the stone pile surfaces in Winter 2026 raises
16 the question of where stone piles will be most effective over time. In sheltered bays, stone
17 piles are protected from wave damage, but algal growth and sediment accumulation could
18 progressively reduce crevice accessibility and refugia value. In more wave- or current-exposed
19 locations, natural scouring could limit fouling, but conditions may be too disturbed to provide
20 effective shelter for kōura ([Parkyn & Collier, 2004](#)). This placement trade-off warrants direct
21 investigation in future deployments. Standardised photography of consistent stone faces each
22 monitoring visit would allow fouling development to be tracked alongside changes in kōura
23 use.

24 The five stone pile structures deployed in this study cannot be expected to drive population-level
25 recovery across a lake of this size. Even if each structure functions as an effective refuge for
26 kōura, population-level recovery requires habitat enhancement at a spatial scale well beyond
27 the current deployment. The spatial and temporal scales of ecosystem responses are typically
28 greater than the scale of the interventions used to produce them ([Minns et al., 1996](#)), and scaling
29 from individual stone pile deployment to habitat addition at population-relevant extents is a
30 central challenge in restoration ecology ([Ngwenya et al., 2025](#)). The temporal dimension of this
31 mismatch is equally important as continued monitoring for at least three years is recommended,
32 as the minimum at which the current observed trajectory would become statistically detectable,
33 and to determine whether the Winter 2026 decline was a one-time event. It is expected that
1 kōura abundance at stone pile sites will continue to build over subsequent spring and summer
2 seasons.

3 Beyond the ecological findings, this study established a practical foundation for scaling habitat
4 enhancement in these lakes. The necessary approvals, consents, and construction methodology
5 for deploying stone piles or rocky structures have now been documented and tested, providing
6 a repeatable framework ready for wider application. This study also provided evidence that
7 the structures are unlikely to provide meaningful additional benefit to catfish, addressing a

8 key management concern before any wider deployment. Together, these outcomes give lake
9 managers with a practical, tested option for future large-scale deployment.

10 As thermal stratification and catfish distribution expands under continued warming (Lee et
11 al., 2025; Woolway et al., 2021), the band of suitable habitat available to kōura will likely
12 continue to narrow. In this context, the value of structural refugia is expected to increase
13 over time rather than diminish. Stone pile deployment alone cannot reverse the trajectory of
14 kōura decline in Lake Rotoiti, but it directly addresses the current bottleneck, the shortage
15 of structural refuge in a lake where soft sediment is dominant on the shorelines and the two
16 primary stressors cannot be controlled at the lake scale. Establishing refugia before populations
17 decline further may be more effective than waiting until the effect of the stressors becomes
18 more severe and kōura source populations fall below critical densities for recolonisation (I. A.
19 K. Kusabs et al., 2026; Morelli et al., 2016).

20 The consistent positive directional trend in occupancy, the rapid seasonal use of the structures
21 documented here, and the developing community on the stone pile surfaces all suggest the
22 foundations for a population-level response by kōura are being established. Furthermore, stone-
23 pile construction provided no evidence of attracting or concentrating catfish. Whether these
24 responses ultimately translate into detectable population-level effects will require continued
25 long-term monitoring.

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1 **Authors' contributions**

2 Study conception and design were undertaken by Olivier V. Raven, Ian A. K. Kusabs, Robin
3 Holmes, Francis J. Burdon and Deniz Özkundakci. Fieldwork was executed by Olivier V. Raven.
4 Data analysis was performed by Olivier V. Raven. Ian A. K. Kusabs obtained council and iwi
1 consent and provided local ecological knowledge. Manuscript drafting was led by Olivier V.
2 Raven. All authors contributed to manuscript revision, approved the results, and the final
3 manuscript.

4 **Use of Artificial Intelligence**

5 Use of Artificial Intelligence: Claude (Anthropic) was used to assist with manuscript editing and
1 language refinement. All content was critically reviewed, and the authors take full responsibility
2 for the final manuscript.

3 **Declarations**

4 The corresponding author has declared that none of the authors has any competing interests.

5 **Data availability statement**

6 The code and derived data supporting the findings of this study are openly available on GitHub
7 (https://github.com/OlivierRaven/Stone_Piles_Rotoiti) and a rendered version of the analysis
8 is hosted at https://olivierraven.github.io/Stone_Piles_Rotoiti/. The primary raw dataset is
9 archived on Zenodo (<https://doi.org/...>). Peer review documents, including the cover letter
10 and reviewer responses, are available in the manuscript repository in the interest of open and
11 transparent science.

12 Tables

Table 1: Physical characteristics of stone pile, control, and reference monitoring sites at baseline. Sites 1–10 are stone pile (odd) and paired control (even) sites; sites 11–20 are reference sites. Substrate index is a weighted composite of substrate particle sizes (higher values indicate coarser substrate more suitable for kōura). Slope is the gradient to the 20 m depth contour. Fetch is maximum effective fetch. Vegetation values are visually estimated percentage cover.

Site	Type	Habitat	Substrate index	Slope (m/m)	Fetch (km)	Riparian veg. (%)	Overhang (%)	Wood cover (%)
1	Stone pile	Rocky	5.20	0.12	4.76	0	10	0
2	Control	Rocky	4.40	0.11	4.79	70	0	0
3	Stone pile	Sandy	3.00	0.01	1.71	100	0	0
4	Control	Sandy	3.00	0.01	1.74	80	5	0
5	Stone pile	Sandy	2.40	0.01	0.61	10	0	0
6	Control	Sandy	3.00	0.01	0.55	0	0	0
7	Stone pile	Sandy	3.00	0.06	2.50	100	0	0
8	Control	Sandy	3.00	0.05	2.22	30	0	0
9	Stone pile	Sandy	3.35	0.08	3.59	0	0	3
10	Control	Rocky	5.00	0.09	3.74	20	0	0
11	Reference	Rocky	6.60	0.16	4.47	0	0	0
12	Reference	Rocky	6.20	0.12	4.17	100	100	0
13	Reference	Sandy	3.60	0.09	1.71	100	100	0
14	Reference	Sandy	3.50	0.11	1.65	100	100	0
15	Reference	Rocky	5.20	0.03	1.44	100	100	6
16	Reference	Rocky	5.20	0.04	1.40	100	100	2
17	Reference	Rocky	5.00	0.05	3.47	100	100	30
18	Reference	Rocky	6.20	0.05	3.46	100	100	7
19	Reference	Rocky	5.70	0.11	2.63	100	100	4
20	Reference	Rocky	5.70	0.13	2.87	100	100	0

13 Source: [Article Notebook](#)

14 Figures

15 Supplementary information

16 *Supplementary figures are provided in the appendices.*



Figure 1: Map of Lake Rotoiti in Bay of Plenty region of New Zealand. The monitoring sites are coloured by site type: stone pile sites red, control site yellow, and reference sites blue. Ten close-up maps show the placement of each of the sites.

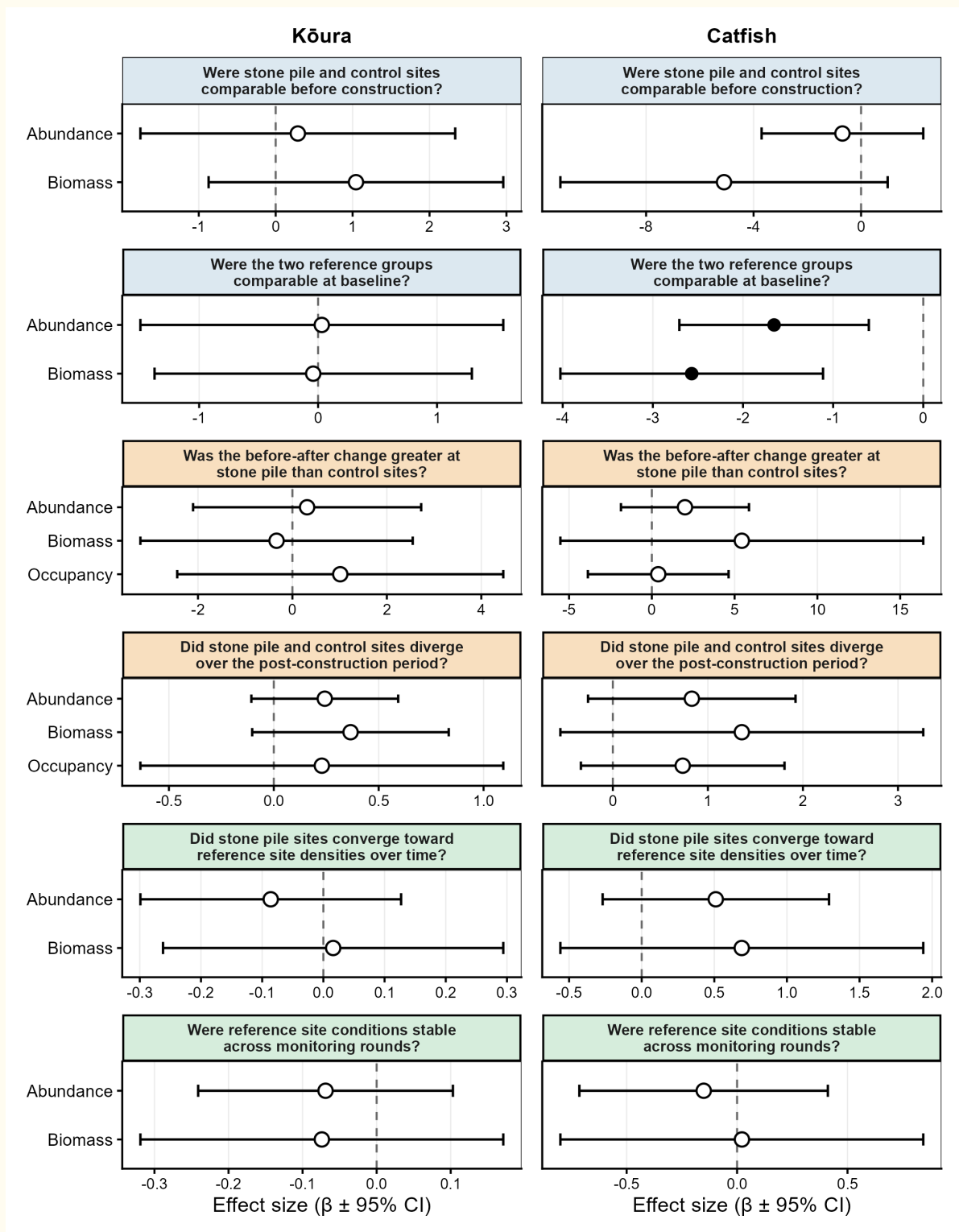


Figure 2: Effect sizes ($\beta \pm 95\% \text{ CI}$) for the key term of interest across all models, separated by species (columns) and comparison type (rows). Filled points indicate significant effects ($p < 0.05$); open points are non-significant. Each row has its own x-axis scale, effect magnitudes are not directly comparable across rows because each comparison tests a different contrast (step-change, per-round slope, or simple group difference), and occupancy is on the log-odds scale while abundance and biomass models use the Tweedie log link. Models with degenerate confidence intervals are excluded.

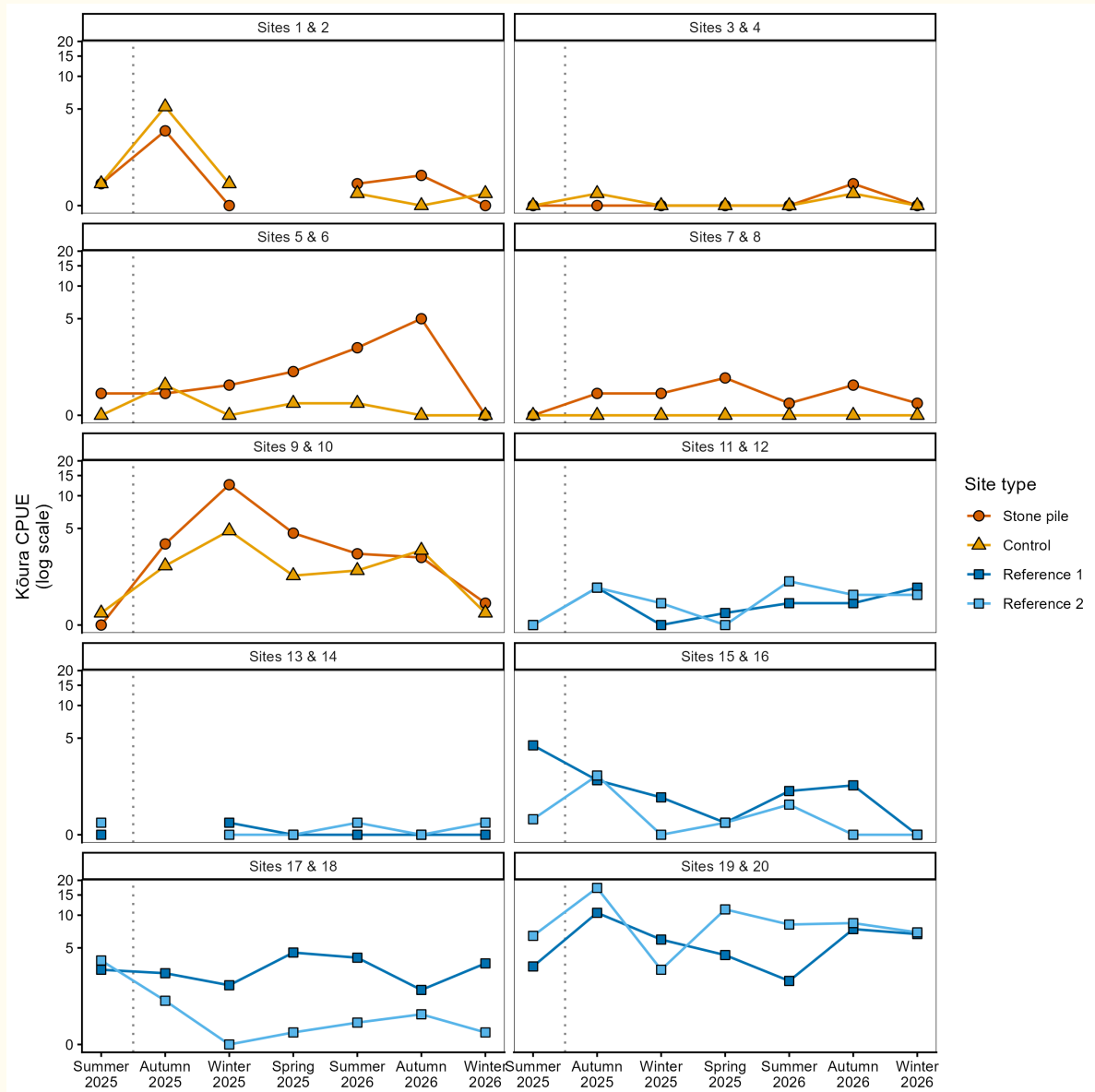


Figure 3: Kōura CPUE at individual site pairs across all monitoring rounds. Each panel shows one stone pile / control pair (sites 1–10) or reference pair (sites 11–20). Colour and shape indicate site type. Vertical dotted line marks stone pile construction. y-axis on $\log(1+x)$ scale.

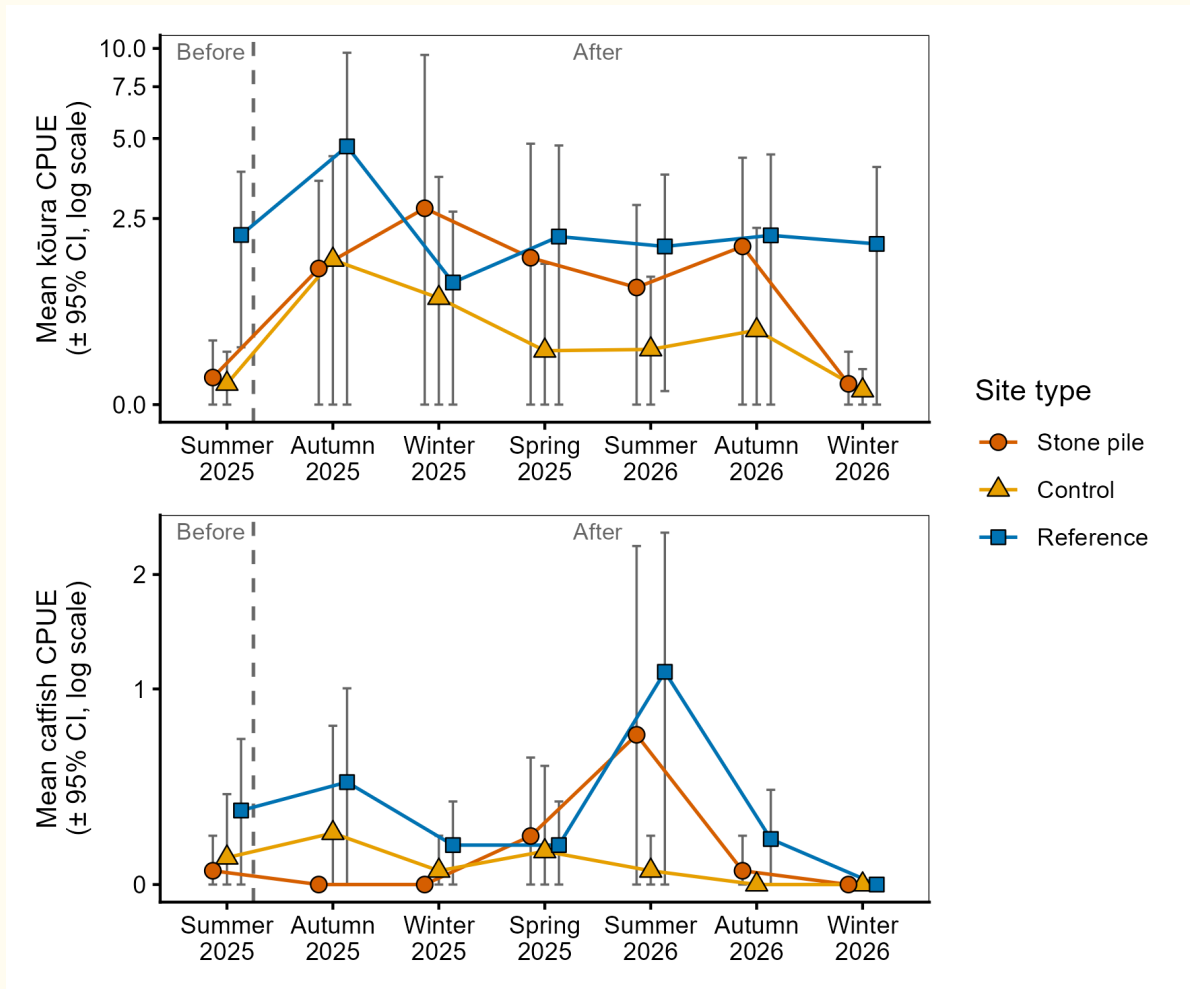


Figure 4: Mean CPUE per monitoring round ($\pm 95\%$ CI, log scale) by site type for kōura (top) and catfish (bottom). Vertical dashed line distinguishes the pre-construction period (Before) from post-construction monitoring (After).

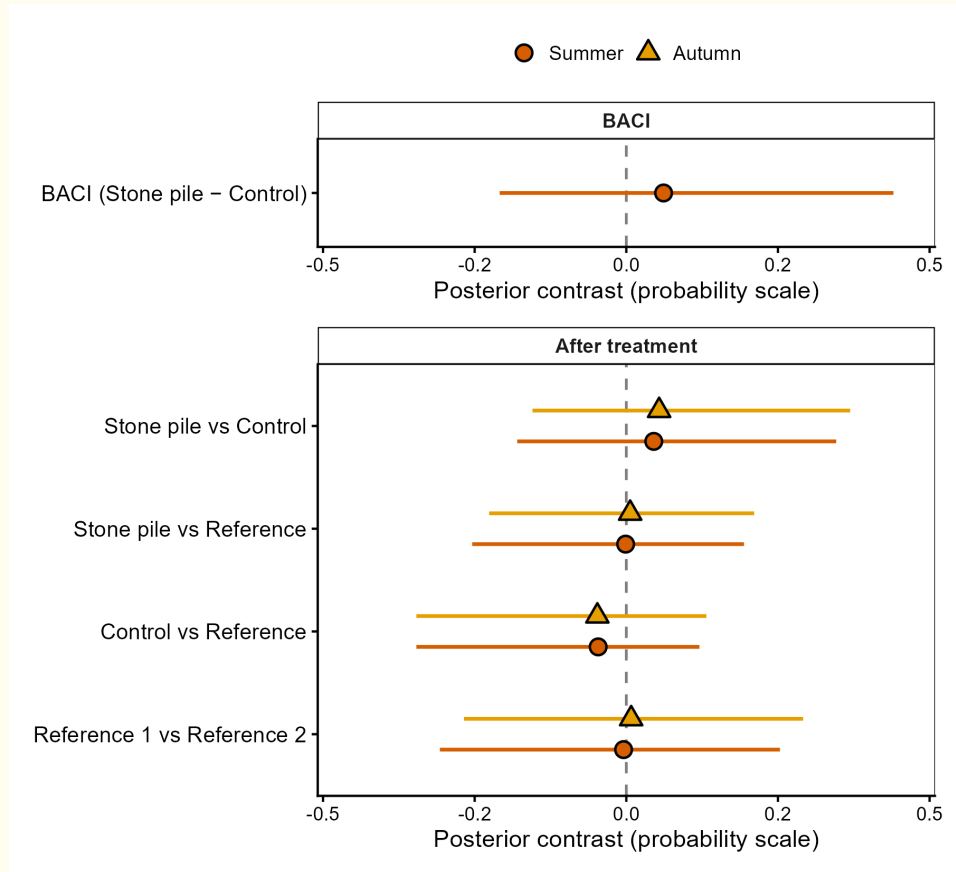


Figure 5: Posterior contrasts in kōura occupancy from the warm-season model. BACI contrast (stone pile minus control, before vs after construction). After-treatment contrasts between all site-type pairs. Points show posterior means with 95 % credible intervals; contrasts overlapping zero indicate no detectable difference.

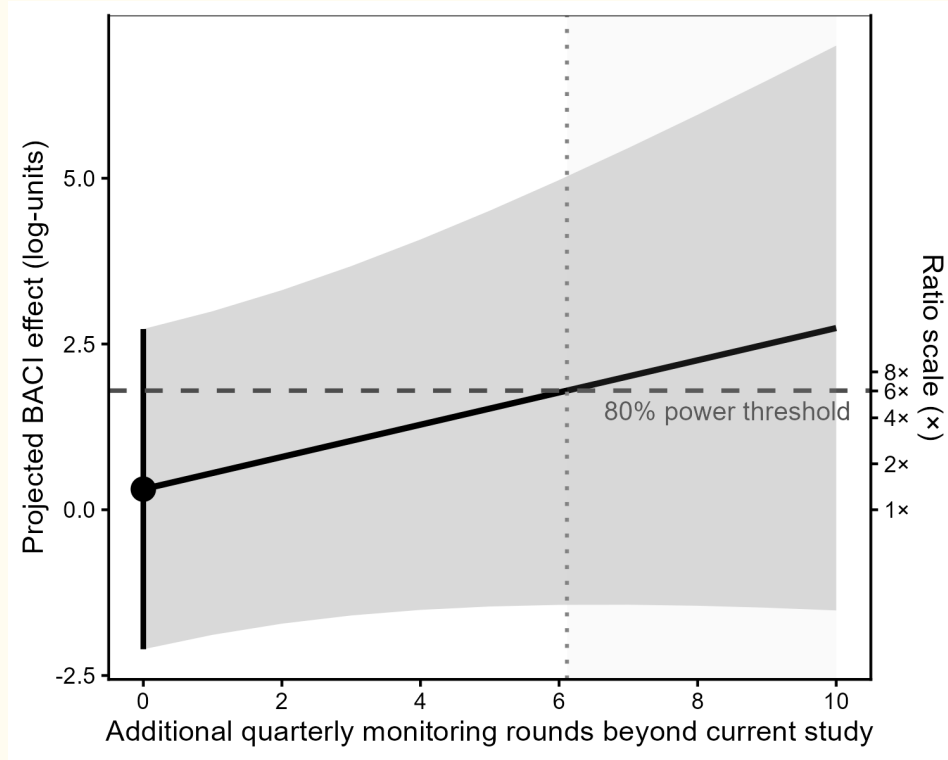


Figure 6: Projected kōura CPUE BACI effect size over additional monitoring rounds. The solid line shows the linear projection based on the estimated post-construction trajectory; shaded ribbon shows 95% uncertainty interval. Dashed horizontal line marks the detection threshold for 80 % power with five site pairs.

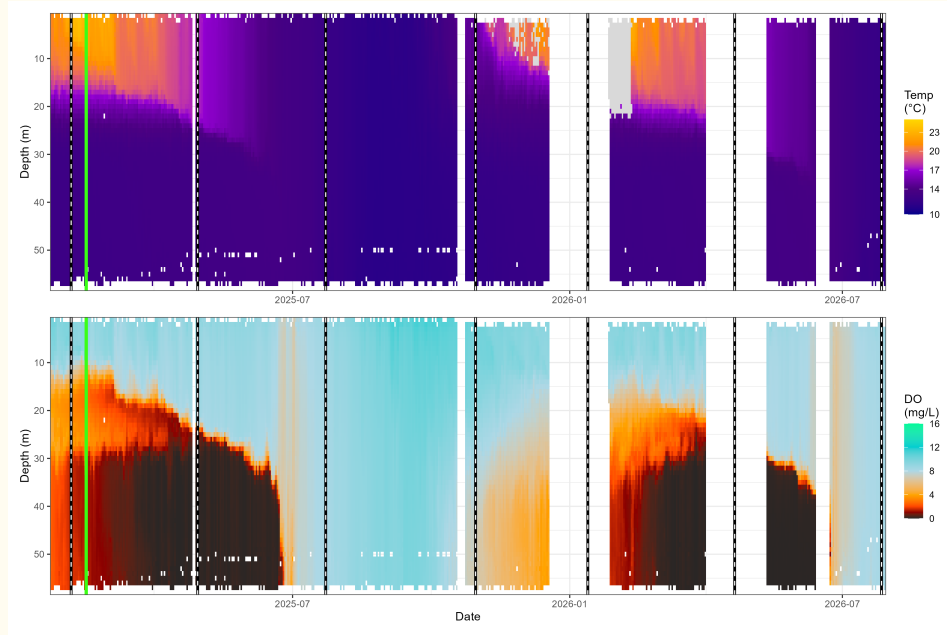


Figure 7: Depth-resolved water temperature ($^{\circ}\text{C}$, top) and dissolved oxygen (mg/L , bottom) in Lake Rotoiti throughout the study period (February 2025 – July 2026). Data derived from the LimnoTrack continuous monitoring buoy ([Moore & McBride, 2026](#)). Vertical green line marks stone pile deployment; dashed vertical lines mark seasonal monitoring rounds. Warm colours in the DO panel (orange–black) indicate hypoxic and anoxic conditions ($< 5 \text{ mg/L}$), demonstrating the pronounced summer stratification that compresses oxygenated habitat into the shallow littoral zone.

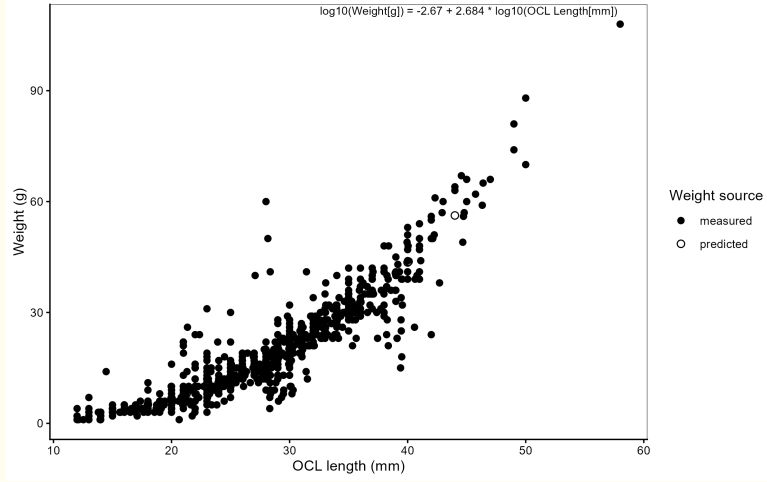


Figure 8: Length–weight relationship for kōura, fitted on log –log scale. Open points are weights predicted from the length–weight regression (used to fill missing scale measurements); filled points are directly measured weights.

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

model	RQ	species	re- sponse	com- pari- type	time_ frame	term	es- ti- mate	std. error	statis- tic	p.value	adj. p	BF	low	high
M1a	a	RQ1	Cat-	abun-	treat-	be-	(Intercept)	-	0.908	-	0.011	NA	-	-
		fish	dance	ment	fore			2.303	2.535			4.083	0.522	
M1a	a	RQ1	Cat-	abun-	treat-	be-	Site_type_de-	-	1.535	-	0.652	0.783	-	2.315
		fish	dance	ment	fore		tailStone_pile	0.693	0.452			3.701		
M1a	b	RQ1	Cat-	biomas-	treat-	be-	(Intercept)	3.203	1.279	2.504	0.012	NA	0.696	5.709
		fish		ment	fore									
M1a	b	RQ1	Cat-	biomas-	treat-	be-	Site_type_de-	-	3.107	-	0.101	0.403	-	0.990
		fish		ment	fore		tailStone_pile	5.100	1.641			11.190		
M1a	a	RQ1	Kōura	abun-	treat-	be-	(Intercept)	-	0.785	-	0.016	NA	-	-
			dance	ment	fore			1.897	2.415			3.437	0.358	
M1a	a	RQ1	Kōura	abun-	treat-	be-	Site_type_de-	0.288	1.044	0.276	0.783	0.783	-	2.334
			dance	ment	fore		tailStone_pile					1.759		
M1a	b	RQ1	Kōura	biomas-	treat-	be-	(Intercept)	1.030	0.830	1.241	0.215	NA	-	2.656
				ment	fore							0.597		

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

model	re-	com-	es-	ti-	std.er-	statis-	p.value	adjco	BFlow	low	high		
Q1	species	response	type	time	frame	mate	ror	tic	p.value	adjco	BFlow	low	high
M1ab	RQ1	Kōura	biomass	treat-	be-	Site_type_de-	1.044	0.977	1.068	0.286	0.571	-	2.959
			ment	fore	tail	Stone_pile					0.872		
M1ba	RQ1	Cat-	abun-	ref-	be-	(Intercept)	-	32.656	-	0.002	NA	-	-
		fish	dance	er-	fore		102.299	3.133			166.308	38.295	
			ence										
M1ba	RQ1	Cat-	abun-	ref-	be-	Site_type_de-	-	0.536	-	0.002	0.004	-	-
		fish	dance	er-	fore	tail	Reference_2	1.655	3.087		2.706	0.604	
			ence										
M1ba	RQ1	Cat-	abun-	ref-	be-	Temperature	4.079	1.293	3.154	0.002	NA	1.544	6.613
		fish	dance	er-	fore								
			ence										
M1bb	RQ1	Cat-	biomass	ref-	be-	(Intercept)	-	37.363	-	0.000	NA	-	-
		fish		er-	fore		153.136	4.099			226.367	79.906	
			ence										
M1bb	RQ1	Cat-	biomass	ref-	be-	Site_type_de-	-	0.743	-	0.001	0.002	-	-
		fish		er-	fore	tail	Reference_2	2.569	3.457		4.026	1.112	
			ence										
M1bb	RQ1	Cat-	biomass	ref-	be-	Temperature	6.315	1.482	4.261	0.000	NA	3.410	9.220
		fish		er-	fore								
			ence										
M1ba	RQ1	Kōura	abun-	ref-	be-	(Intercept)	0.742	0.553	1.343	0.179	NA	-	1.825
			dance	er-	fore						0.341		
			ence										
M1ba	RQ1	Kōura	abun-	ref-	be-	Site_type_de-	0.031	0.778	0.040	0.968	0.968	-	1.557
			dance	er-	fore	tail	Reference_2				1.494		
			ence										
M1bb	RQ1	Kōura	biomass	ref-	be-	(Intercept)	3.623	0.478	7.580	0.000	NA	2.686	4.560
				er-	fore								
			ence										

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

model	Re- Q1	re- species	com- pari- response_type	time_frame	es- ti- std.er-statistic	p.value	adjco	BFlow	conf.high
M1bb	RQ1K	our	biomassref- er- ence	be- fore tail	Site_type_de- Reference_2	- 0.681 0.041	- 0.952 0.061	20.968 1.375	- 1.293
M2aad	RQ2C	Cat- fish	abun- dance	vs_confull trol	(Intercept)	- 1.631 3.976	- 0.015 2.438	NA 7.173	- 0.780
M2aad	RQ2C	Cat- fish	abun- dance	vs_confull trol	PeriodAfter	0.673 1.521	0.442 0.658	NA 2.308	- 3.654
M2aad	RQ2C	Cat- fish	abun- dance	vs_confull trol	siteStone_pile	- 1.762 0.639	- 0.717 0.362	NA 4.093	- 2.816
M2aad	RQ2C	Cat- fish	abun- dance	vs_confull trol	monitoring_int	- 0.344 0.294	- 0.394 0.853	NA 0.968	- 0.381
M2aad	RQ2C	Cat- fish	abun- dance	vs_confull trol	fetch_z	- 1.077 1.194	- 0.267 1.109	NA 3.305	- 0.916
M2aad	RQ2C	Cat- fish	abun- dance	vs_confull trol	Peri- odAfter:site- Stone_pile	2.002 1.973	1.015 0.310	0.858 1.864	- 5.869
M2abb	RQ2C	Cat- fish	biomassvs- trol	vs_confull trol	(Intercept)	0.202 4.319	0.047 0.963	NA 8.262	- 8.667
M2abb	RQ2C	Cat- fish	biomassvs- trol	vs_confull trol	PeriodAfter	- 2.404 0.340	- 0.888 0.141	NA 5.052	- 4.372
M2abb	RQ2C	Cat- fish	biomassvs- trol	vs_confull trol	siteStone_pile	- 7.158 5.574	- 0.436 0.779	NA 19.604	- 8.456
M2abb	RQ2C	Cat- fish	biomassvs- trol	vs_confull trol	monitoring_int	- 0.523 0.381	- 0.466 0.728	NA 1.406	- 0.644
M2abb	RQ2C	Cat- fish	biomassvs- trol	vs_confull trol	fetch_z	- 2.438 1.675	- 0.492 0.687	NA 6.454	- 3.103
M2abb	RQ2C	Cat- fish	biomassvs- trol	vs_confull trol	Peri- odAfter:site- Stone_pile	5.436 5.592	0.972 0.331	0.858 5.524	- 16.396

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

model	Q1	species	re- response	com- pari- type	time	frame	es- ti- mate	std.er	statis- tic	p.value	adjcoB	Flow	low	high
M2aod	RQ26	Cat-	occu-	vs_confull		(Intercept)	-	1.887	-	0.247	NA	-	1.513	
		fish	pancy	trol			2.185		1.158			5.883		
M2aod	RQ26	Cat-	occu-	vs_confull		PeriodAfter	-	1.765	-	0.927	NA	-	3.299	
		fish	pancy	trol			0.161		0.091			3.621		
M2aod	RQ26	Cat-	occu-	vs_confull		siteStone_pile	0.024	1.982	0.012	0.990	NA	-	3.909	
		fish	pancy	trol								3.860		
M2aod	RQ26	Cat-	occu-	vs_confull		monitoring_int	-	0.259	-	0.525	NA	-	0.343	
		fish	pancy	trol			0.165		0.635			0.673		
M2aod	RQ26	Cat-	occu-	vs_confull		fetch_z	-	1.197	-	0.423	NA	-	1.387	
		fish	pancy	trol			0.960		0.802			3.306		
M2aod	RQ26	Cat-	occu-	vs_confull		Peri-	0.387	2.168	0.179	0.858	0.858	-	4.637	
		fish	pancy	trol		odAfter:site-						3.863		
						Stone_pile								
M2aa	RQ2K	Kōura	abun-	vs_confull		(Intercept)	-	1.183	-	0.020	NA	-	-	
		dance	trol				2.753		2.327			5.072	0.434	
M2aa	RQ2K	Kōura	abun-	vs_confull		PeriodAfter	2.374	0.976	2.432	0.015	NA	0.461	4.286	
		dance	trol											
M2aa	RQ2K	Kōura	abun-	vs_confull		siteStone_pile	0.751	1.335	0.563	0.574	NA	-	3.368	
		dance	trol									1.865		
M2aa	RQ2K	Kōura	abun-	vs_confull		monitoring_int	-	0.082	-	0.002	NA	-	-	
		dance	trol				0.253		3.094			0.413	0.093	
M2aa	RQ2K	Kōura	abun-	vs_confull		Peri-	0.311	1.232	0.253	0.800	0.858	-	2.727	
		dance	trol			odAfter:site-						2.104		
						Stone_pile								
M2ab	RQ2K	Kōura	biomass	vs_confull		(Intercept)	0.315	1.335	0.236	0.813	NA	-	2.931	
			trol									2.301		
M2ab	RQ2K	Kōura	biomass	vs_confull		PeriodAfter	2.248	1.246	1.805	0.071	NA	-	4.689	
			trol									0.193		
M2ab	RQ2K	Kōura	biomass	vs_confull		siteStone_pile	1.410	1.552	0.909	0.363	NA	-	4.451	
			trol									1.631		

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

model	re- Q1	com- pari- response_type	time_frame	es- ti- mate	std.er- ror	statis- tic	p.value	adjco- Bflow	low	high
M2abRQ2K	Kōura	biomass	vs_confull	monitoring_int	- 0.110	- 0.073	NA	- 0.019		
		trol			0.196	1.791		0.412		
M2abRQ2K	Kōura	biomass	vs_confull	Peri-	- 1.471	- 0.819	0.858	- 2.548		
		trol		odAfter:site-	0.336	0.228		3.220		
				Stone_pile						
M2aoRQ2K	Kōura	occu-	vs_confull	(Intercept)	- 1.643	- 0.674	NA	- 2.530		
		pancy	trol		0.690	0.420		3.911		
M2aoRQ2K	Kōura	occu-	vs_confull	PeriodAfter	1.913	1.556	1.230	0.219	NA	- 4.963
		pancy	trol					1.136		
M2aoRQ2K	Kōura	occu-	vs_confull	siteStone_pile	0.013	2.121	0.006	0.995	NA	- 4.171
		pancy	trol					4.144		
M2aoRQ2K	Kōura	occu-	vs_confull	monitoring_int	- 0.215	- 0.183	NA	- 0.135		
		pancy	trol		0.286	1.333		0.708		
M2aoRQ2K	Kōura	occu-	vs_confull	Peri-	1.013	1.762	0.575	0.565	0.858	- 4.466
		pancy	trol	odAfter:site-				2.440		
				Stone_pile						
M2baRQ2C	Cat-	abun-	vs_conaf-	(Intercept)	- 1.519	- 0.157	NA	- 0.826		
	fish	dance	trol		2.150	1.416		5.127		
M2baRQ2C	Cat-	abun-	vs_conaf-	siteStone_pile	- 1.832	- 0.447	NA	- 2.199		
	fish	dance	trol		1.392	0.760		4.983		
M2baRQ2C	Cat-	abun-	vs_conaf-	monitoring_int	- 0.406	- 0.118	NA	- 0.162		
	fish	dance	trol		0.634	1.562		1.430		
M2baRQ2C	Cat-	abun-	vs_conaf-	fetch_z	- 0.976	- 0.261	NA	- 0.816		
	fish	dance	trol		1.097	1.124		3.010		
M2baRQ2C	Cat-	abun-	vs_conaf-	site-	0.831	0.557	1.492	0.136	0.215	- 1.922
	fish	dance	trol	Stone_pile:mon-				0.261		
				itoring_int						
M2bbRQ2C	Cat-	biomass	vs_conaf-	(Intercept)	2.964	2.725	1.088	0.277	NA	- 8.304
	fish		trol					2.377		

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

model	Re- Q1 species	com- re- pari- response	type	time	frame	es- ti- mate	std.er- ror	statis- tic	p.value	adjco- Bflow	llow	hhigh
M2bbRQ26	Cat- fish	biomass	vs_ trol	conaf- ter	site	Stone_pile	- 4.240	- 0.286	NA	- 3.789		
							4.522	1.066		12.833		
M2bbRQ26	Cat- fish	biomass	vs_ trol	conaf- ter	monitoring_int		- 0.688	- 0.157	NA	- 0.374		
							0.974	1.416		2.323		
M2bbRQ26	Cat- fish	biomass	vs_ trol	conaf- ter	fetch_z		- 1.390	- 0.524	NA	- 1.837		
							0.886	0.638		3.610		
M2bbRQ26	Cat- fish	biomass	vs_ trol	conaf- ter	site- Stone_pile:mon- itoring_int		1.357 0.974 1.393	0.1640.215	- 3.266	0.552		
M2boRQ26	Cat- fish	occu- pancy	vs_ trol	conaf- ter	(Intercept)		- 1.737	- 0.584	NA	- 2.453		
							0.952	0.548		4.357		
M2boRQ26	Cat- fish	occu- pancy	vs_ trol	conaf- ter	site	Stone_pile	- 1.936	- 0.298	NA	- 1.781		
							2.013	1.040		5.808		
M2boRQ26	Cat- fish	occu- pancy	vs_ trol	conaf- ter	monitoring_int		- 0.420	- 0.179	NA	- 0.258		
							0.565	1.345		1.388		
M2boRQ26	Cat- fish	occu- pancy	vs_ trol	conaf- ter	fetch_z		- 1.121	- 0.444	NA	- 1.339		
							0.858	0.766		3.055		
M2boRQ26	Cat- fish	occu- pancy	vs_ trol	conaf- ter	site- Stone_pile:mon- itoring_int		0.735 0.546 1.344	0.1790.215	- 1.806	0.336		
M2baRQ2K	Kōura	abun- dance	vs_ trol	conaf- ter	(Intercept)		0.084 0.846 0.099	0.921	NA	- 1.741		
										1.573		
M2baRQ2K	Kōura	abun- dance	vs_ trol	conaf- ter	site	Stone_pile	0.323 0.808 0.400	0.689	NA	- 1.907		
										1.261		
M2baRQ2K	Kōura	abun- dance	vs_ trol	conaf- ter	monitoring_int		- 0.143	- 0.004	NA	- -		
							0.409	2.861		0.690 0.129		
M2baRQ2K	Kōura	abun- dance	vs_ trol	conaf- ter	site- Stone_pile:mon- itoring_int		0.243 0.179 1.359	0.1740.215	- 0.593	0.107		

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

model	Q1	species	response	com- re- pari- type	time_frame	es- ti- mate	std.er- ror	statis- tic	p.value	adjco- Bflow	low	high
M2bb	RQ2K	Kōura	biomass	vs_conaf- trol ter	(Intercept)	3.270	0.856	3.821	0.000	NA	1.593	4.948
M2bb	RQ2K	Kōura	biomass	vs_conaf- trol ter	siteStone_pile	-	0.986	-	0.938	NA	-	1.856
						0.077		0.078			2.010	
M2bb	RQ2K	Kōura	biomass	vs_conaf- trol ter	monitoring_int	-	0.191	-	0.025	NA	-	-
						0.427		2.240			0.800	0.053
M2bb	RQ2K	Kōura	biomass	vs_conaf- trol ter	site- Stone_pile:mon- itoring_int	0.366	0.239	1.531	0.126	0.215	-	0.834
											0.103	
M2bo	RQ2K	Kōura	occu- pancy	vs_conaf- trol ter	(Intercept)	1.585	1.885	0.841	0.400	NA	-	5.280
											2.109	
M2bo	RQ2K	Kōura	occu- pancy	vs_conaf- trol ter	siteStone_pile	0.563	2.660	0.212	0.832	NA	-	5.776
											4.651	
M2bo	RQ2K	Kōura	occu- pancy	vs_conaf- trol ter	monitoring_int	-	0.311	-	0.195	NA	-	0.207
						0.403		1.296			1.014	
M2bo	RQ2K	Kōura	occu- pancy	vs_conaf- trol ter	site- Stone_pile:mon- itoring_int	0.229	0.441	0.519	0.604	0.604	-	1.094
											0.636	
M3aad	RQ3C	Cat- fish	abun- dance	vs_ref-af- er- ence	(Intercept)	-	1.282	-	0.000	NA	-	-
						7.151		5.577			9.664	4.638
M3aad	RQ3C	Cat- fish	abun- dance	vs_ref-af- er- ence	siteStone_pile	-	1.940	-	0.203	NA	-	1.331
						2.471		1.274			6.273	
M3aad	RQ3C	Cat- fish	abun- dance	vs_ref-af- er- ence	monitoring_int	-	0.147	-	0.144	NA	-	0.073
						0.215		1.462			0.504	
M3aad	RQ3C	Cat- fish	abun- dance	vs_ref-af- er- ence	Temperature	0.315	0.058	5.414	0.000	NA	0.201	0.429

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

model	Re- Q1 species	com- re- pari- response_type	time_frame	es- ti- std.er-statis- mate ror tic	p.value	adjcoBFlow	low	high
M3aaRQ3C	Cat- fish	abun- dance	vs_ref-af- er- ence	site- Stone_pile:mon- itoring_int	0.510 0.398 1.283 0.199	0.559	-	1.290
M3abRQ3C	Cat- fish	biomass	vs_ref-af- er- ence	(Intercept)	3.011 1.189 2.532 0.011	NA	0.681	5.342
M3abRQ3C	Cat- fish	biomass	vs_ref-af- er- ence	siteStone_pile	- 3.059 - 0.153	NA	-	1.626
M3abRQ3C	Cat- fish	biomass	vs_ref-af- er- ence	monitoring_int	- 0.209 - 0.232	NA	-	0.160
M3abRQ3C	Cat- fish	biomass	vs_ref-af- er- ence	(Intercept)	0.229 0.519 0.440 0.660	NA	-	1.247
M3aaRQ3K	Kōura- abun- dance	vs_ref-af- er- ence	siteStone_pile	0.217 0.901 0.240 0.810	NA	-	1.984	
M3aaRQ3K	Kōura- abun- dance	vs_ref-af- er- ence	monitoring_int	- 0.056 - 0.107	NA	-	0.019	
M3aaRQ3K	Kōura- abun- dance	vs_ref-af- er- ence	site- Stone_pile:mon- itoring_int	- 0.109 - 0.429	0.572	-	0.127	
M3abRQ3K	Kōura- abun- dance	biomass	vs_ref-af- er- ence	(Intercept)	3.369 0.506 6.655 0.000	NA	2.377	4.361

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

model	Re- sponse	com- pari- son	time	frame	es- ti- mate	std.er- ror	statis- tic	p.value	adjco- eff	Flow	low	high
M3abRQ3K	Kourab	biomass	svs_ref- er- ence	af- ter	siteStone_pile	- 0.885 0.149	- 0.866 0.168	NA	-	1.586 1.884		
M3abRQ3K	Kourab	biomass	svs_ref- er- ence	af- ter	monitoring_int	- 0.074 0.088	- 0.239 1.177	NA	-	0.058 0.234		
M3abRQ3K	Kourab	biomass	svs_ref- er- ence	af- ter	site- Stone_pile:mon- itoring_int	0.016	0.142 0.112	0.911 0.911	-	0.294 0.262		
M3baRQ3K	Cat- fish	abun- dance	ref- er- ence	af- ter	(Intercept)	- 1.705 7.372	- 0.000 4.324	NA	-	- 10.713	-	- 1.031
M3baRQ3K	Cat- fish	abun- dance	ref- er- ence	af- ter	Site_type_de- tailReference_2	0.753	1.007 0.747	0.455 NA	-	2.726 1.221		
M3baRQ3K	Cat- fish	abun- dance	ref- er- ence	af- ter	monitoring_int	- 0.220 0.130	- 0.555 0.591	NA	-	0.301 0.562		
M3baRQ3K	Cat- fish	abun- dance	ref- er- ence	af- ter	Temperature	0.304	0.073 4.183	0.000 NA	0.161	0.446		
M3baRQ3K	Cat- fish	abun- dance	ref- er- ence	af- ter	Site_type_de- tailRefer- ence_2:moni- toring_int	- 0.287 0.151	- 0.598 0.527	0.797	-	0.411 0.714		
M3bbRQ3K	Cat- fish	biomass	ref- er- ence	af- ter	(Intercept)	2.726	1.556 1.752	0.080 NA	-	5.775 0.324		

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

model	Q1	Q2	species	response	com- pari- type	time	frame	es- ti- mate	std.er	statis- tic	p.value	adjco	BFlow	low	high
M3bb	RQ36	Cat-	fish	biomass	ref- er- ence	af- ter	Site_type_de- tailReference_2	0.191	1.640	0.116	0.907	NA	-	3.406	
M3bb	RQ36	Cat-	fish	biomass	ref- er- ence	af- ter	monitoring_int	-	0.301	-	0.451	NA	-	0.363	
								0.227		0.754			0.816		
M3bb	RQ36	Cat-	fish	biomass	ref- er- ence	af- ter	Site_type_de- tailRefer- ence_2;moni- toring_int	0.022	0.419	0.052	0.959	0.959	-	0.843	
													0.800		
M3ba	RQ3K	Kōura	abun-	dance	ref- er- ence	af- ter	(Intercept)	0.380	0.790	0.480	0.631	NA	-	1.928	
													1.169		
M3ba	RQ3K	Kōura	abun-	dance	ref- er- ence	af- ter	Site_type_de- tailReference_2	-	0.570	-	0.840	NA	-	1.002	
								0.115		0.202			1.232		
M3ba	RQ3K	Kōura	abun-	dance	ref- er- ence	af- ter	monitoring_int	-	0.062	-	0.763	NA	-	0.103	
								0.019		0.302			0.141		
M3ba	RQ3K	Kōura	abun-	dance	ref- er- ence	af- ter	SeasonSpring	-	0.227	-	0.089	NA	-	0.059	
								0.386		1.699			0.831		
M3ba	RQ3K	Kōura	abun-	dance	ref- er- ence	af- ter	SeasonSummer	-	0.233	-	0.157	NA	-	0.127	
								0.330		1.416			0.787		
M3ba	RQ3K	Kōura	abun-	dance	ref- er- ence	af- ter	SeasonWinter	-	0.196	-	0.005	NA	-	-	
								0.546		2.781			0.932	0.161	

Table 2: Full fixed-effect estimates (β , SE, 95% CI, p-value) for every term in every model.

model	Re- sponse	com- pari- son	time	frame	es- ti- mate	std.er- ror	statis- tic	p.value	adjco- p	Flow	low	high	
M3ba	RQ3K	Kōura	abun-	ref-	af-	Site_type_de-	-	0.088	-	0.431	0.797	-	0.103
		dance	er-	ence	ter	tailRefer-	0.069		0.787		0.241		
						ence_2:moni-							
						toring_int							
M3bb	RQ3K	Kōura	biomass	ref-	af-	(Intercept)	3.275	0.800	4.095	0.000	NA	1.707	4.842
			er-	ence	ter								
M3bb	RQ3K	Kōura	biomass	ref-	af-	Site_type_de-	-	0.670	-	0.920	NA	-	1.245
			er-	ence	ter	tailReference_2	0.067		0.100		1.379		
M3bb	RQ3K	Kōura	biomass	ref-	af-	monitoring_int	-	0.087	-	0.584	NA	-	0.122
			er-	ence	ter		0.047		0.548		0.217		
M3bb	RQ3K	Kōura	biomass	ref-	af-	Site_type_de-	-	0.125	-	0.554	0.797	-	0.171
			er-	ence	ter	tailRefer-	0.074		0.591		0.319		
						ence_2:moni-							
						toring_int							

Table 3: Baseline (pre-construction) comparability of stone pile, control, and reference sites. Values are mean $\pm$ SD; p-values test Stone pile vs Control (paired Wilcoxon signed-rank test) and Stone pile vs Reference (unpaired Wilcoxon rank-sum test).

Variable	Stone_pile	Control	Reference	p_vs_Con- trol	p_vs_Ref- erence
Kōura CPUE	0.2 $\pm$ 0.27	0.15 $\pm$ 0.22	2.13 $\pm$ 2.32	1.000	0.205
Catfish CPUE	0.05 $\pm$ 0.11	0.1 $\pm$ 0.22	0.3 $\pm$ 0.52	1.000	0.420
Kōura BPUE	7.95 $\pm$ 12.15	2.8 $\pm$ 3.84	36.69 $\pm$ 36.82	0.789	0.206
Catfish BPUE	0.15 $\pm$ 0.34	24.6 $\pm$ 55.01	97.28 $\pm$ 232.15	1.000	0.343

Table 3: Baseline (pre-construction) comparability of stone pile, control, and reference sites. Values are mean $\pm$ SD; p-values test Stone pile vs Control (paired Wilcoxon signed-rank test) and Stone pile vs Reference (unpaired Wilcoxon rank-sum test).

Variable	Stone_pile	Control	Reference	p_vs_Con- trol	p_vs_Ref- erence
Substrate index (before)	3.39 $\pm$ 1.07	3.68 $\pm$ 0.95	5.29 $\pm$ 1.05	0.789	0.012
Substrate index (after)	3.65 $\pm$ 1.12	3.68 $\pm$ 0.95	5.29 $\pm$ 1.05	1.000	0.032
Slope (0–20 m)	0.06 $\pm$ 0.05	0.06 $\pm$ 0.04	0.09 $\pm$ 0.05	0.590	0.245
Temperature ($^{\circ}$ C)	24.74 $\pm$ 1.92	24.78 $\pm$ 1.47	24.48 $\pm$ 0.78	1.000	0.623
Dissolved oxygen (mg/L)	9.85 $\pm$ 1.48	9.63 $\pm$ 1.29	10.03 $\pm$ 0.86	0.418	0.426
Specific conductivity (μ S/cm)	195.44 $\pm$ 70.05	199.8 $\pm$ 78.52	165.45 $\pm$ 8.5	0.281	0.951
pH	7.5 $\pm$ 2.15	7.61 $\pm$ 2.2	8.53 $\pm$ 0.66	1.000	0.426

Table 4: Posterior kōura occupancy probability (ψ) from the warm-season occupancy model (Summer and Autumn rounds). Values are population-level estimates averaged over random site effects, with 95 % credible intervals. Autumn has no Before period as baseline monitoring was conducted in summer only.

Season	Period	Site type	ψ (mean [95% CrI])
Summer	Before	Stone pile	0.81 [0.26–1.00]
Summer	Before	Control	0.82 [0.34–1.00]
Summer	Before	Reference 1	0.86 [0.36–1.00]
Summer	Before	Reference 2	0.89 [0.49–1.00]
Summer	After	Stone pile	0.93 [0.64–1.00]
Summer	After	Control	0.89 [0.53–1.00]
Summer	After	Reference 1	0.93 [0.62–1.00]
Summer	After	Reference 2	0.94 [0.67–1.00]
Autumn	After	Stone pile	0.93 [0.69–1.00]
Autumn	After	Control	0.88 [0.53–1.00]
Autumn	After	Reference 1	0.93 [0.65–1.00]
Autumn	After	Reference 2	0.92 [0.63–1.00]

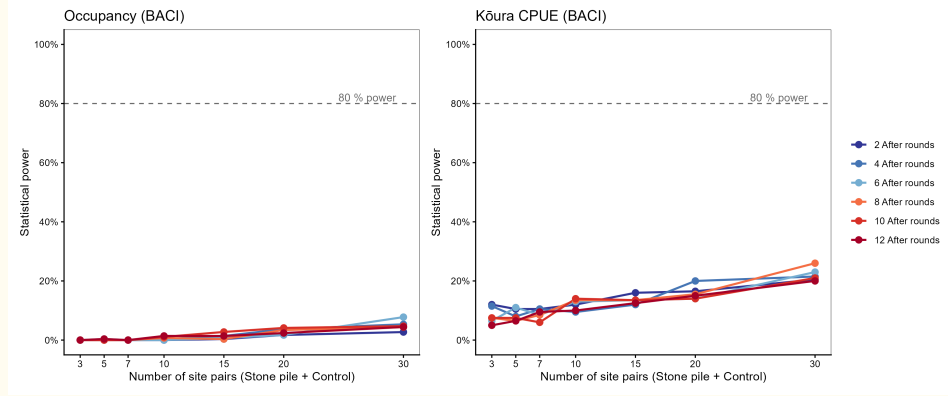


Figure 9: Simulation-based power curves for the occupancy BACI (left) and kōura CPUE BACI (right). Each line represents a different number of After-period monitoring rounds. Dashed line marks 80 % power.

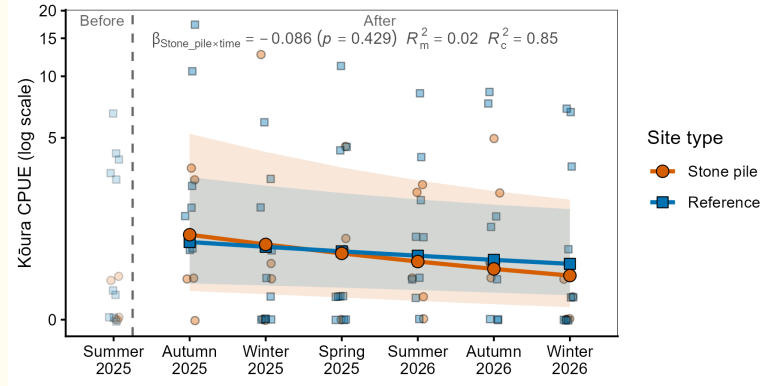


Figure 10: Predicted kōura CPUE trajectories over monitoring rounds at stone pile and reference sites from the abundance convergence model. Shaded ribbons represent 95% confidence intervals. The annotation shows the key interaction term ($\beta \pm p\text{-value}$) and model R^2 values for fixed effects (R^2_m) and conditional (R^2_c).

- Adams, S. B. (2007). Direct and indirect effects of channel catfish (*Ictalurus punctatus*) on native crayfishes (Cambaridae) in experimental tanks. *The American Midland Naturalist*, 158(1), 85–96. [https://doi.org/10.1674/0003-0031\(2007\)158%5B85:daieoc%5D2.0.co;2](https://doi.org/10.1674/0003-0031(2007)158%5B85:daieoc%5D2.0.co;2)
- Barnes, G. E. (1996). *The biology and general ecology of the brown bullhead catfish (ameiurus nebulosus) in Lake Taupo* [PhD thesis]. University of Waikato.
- Barnes, G. E., & Hicks, B. J. (2003). Brown bullhead catfish (*Ameiurus nebulosus*) in Lake Taupo. *Managing Invasive Freshwater Fish in New Zealand: Proceedings of a Workshop Hosted by Department of Conservation, 10-12 May 2001, Hamilton*.
- Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1), 1–48. <https://doi.org/10.18637/jss.v067.i01>
- Blanchfield, P. J., Rodrigues, T. H., MacNeil, C., Stewart, S. D., Lee, F., Hicks, B. J., Grayling, S., Fort-D’Ath, S., Prentice, M. J., Baker, C., Williams, P., & Holmes, R. J. P. (2026). Invading brown bullhead catfish (*ameiurus nebulosus*) show thermal niche rigidity and benthic niche plasticity: Implications for management. *Biological Invasions*, 28(9). <https://doi.org/10.1007/s10530-026-03877-5>
- Bonacina, L., Fasano, F., Mezzanotte, V., & Fornaroli, R. (2023). Effects of water temperature on freshwater macroinvertebrates: A systematic review. *Biological Reviews*, 98(1), 191–221. <https://doi.org/10.1111/brv.12903>
- Brooks, M., Bolker, B., Kristensen, K., Maechler, M., Magnusson, A., Skaug, H., Nielsen, A., Berg, C., & Benthani, K. van. (2017). glmmTMB: Generalized linear mixed models using template model builder. In *CRAN: Contributed Packages*. <https://doi.org/10.32614/cran.package.glmmTMB>
- Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software*, 80(1). <https://doi.org/10.18637/jss.v080.i01>
- Bürkner, P.-C. (2018). Advanced Bayesian multilevel modeling with the R package brms. *The R Journal*, 10(1), 395. <https://doi.org/10.32614/rj-2018-017>
- Burnham, K. P., & Anderson, D. R. (2002). Advanced issues and deeper insights. In *Model selection and multimodel inference: A practical information-theoretic approach* (pp. 267–351). Springer. https://doi.org/10.1007/978-0-387-22456-5_6
- Collier, K. J., Parkyn, S. M., & Rabeni, C. F. (1997). Koura: A keystone species. *Water and Atmosphere*, 5(1), 18–20.
- Dedual, M. (2002). Vertical distribution and movements of brown bullhead (*Ameiurus nebulosus* Lesueur 1819) in Motuoapa Bay, southern Lake Taupo, New Zealand. *Hydrobiologia*, 483, 129–135. https://doi.org/10.1007/978-94-017-0771-8_15

- 17 Dedual, M. (2019). *Summary of the impacts and control methods of brown bullhead catfish*
 18 *(Ameiurus nebulosus Lesueur 1815) in New Zealand and overseas*. Bay of Plenty Regional
 19 Council. <https://atlas.boprc.govt.nz/api/v1/edms/document/A3591759/content>
- 20 Devcich, A. A. (1979). *An ecological study of Paranephrops planifrons White (Decapoda :
 21 Parastacidae) in Lake Rotoiti, North Island* [PhD thesis]. University of Waikato.
- 22 Francis, L. B. (2019). *Evaluating the effects of invasive brown bullhead catfish (Ameiurus
 23 nebulosus) on kōura (freshwater crayfish, Paranephrops planifrons) in Lake Rotoiti* [Master's
 24 thesis, University of Waikato]. [https://researchcommons.waikato.ac.nz/handle/10289/
 25 12746](https://researchcommons.waikato.ac.nz/handle/10289/12746)
- 26 Froese, R. (2006). Cube law, condition factor and weight-length relationships: History,
 27 meta-analysis and recommendations. *Journal of Applied Ichthyology*, 22(4), 241–253.
 28 <https://doi.org/10.1111/j.1439-0426.2006.00805.x>
- 29 Gabry, J., Češnovar, R., Johnson, A., & Bronder, S. (2025). *Cmdstanr: R interface to
 30 'CmdStan'*. <https://mc-stan.org/cmdstanr/>
- 31 Grayling, S. (2020). *Bay of Plenty regional pest management plan 2011-2016 annual report
 32 for 2019/2020*. Bay of Plenty Regional Council. [https://atlas.boprc.govt.nz/api/v1/edms/
 33 document/A3651696/content](https://atlas.boprc.govt.nz/api/v1/edms/document/A3651696/content)
- 34 Hamilton, D., McBride, C., & Uraoka, T. (2005). *Lake Rotoiti fieldwork and modelling to
 35 support considerations of Ohau channel diversion from Lake Rotoiti*. NIWA & University of
 1 Waikato.
- 2 Hanski, I. (1998). Metapopulation dynamics. *Nature*, 396, 41–49. [https://doi.org/10.1038/
 3 23876](https://doi.org/10.1038/23876)
- 4 Hartig, F. (2016). DHARMA: Residual diagnostics for hierarchical (multi-level / mixed)
 5 regression models. In *CRAN: Contributed Packages*. The R Foundation. [https://doi.org/
 6 10.32614/cran.package.dharma](https://doi.org/10.32614/cran.package.dharma)
- 7 Hiroa, T. R. (1921). Māori food-supplies of Lake Rotorua, with methods of obtaining them,
 8 and usages and customs appertaining thereto. *Transactions and Proceedings of the Royal
 9 Society of New Zealand 1868-1961*, 53, 433–451.
- 10 Huolila, M., Marjomäki, T. J., & Laukkanen, E. (1997). The success of crayfish stocking
 11 in a dredged river with and without artificial shelter increase. *Fisheries Research*, 32(2),
 12 185–189. [https://doi.org/10.1016/s0165-7836\(97\)00046-5](https://doi.org/10.1016/s0165-7836(97)00046-5)
- 13 Johnsen, S. I., & Taugbøl, T. (2008). Add stones, get crayfish – is it that simple? *Freshwater
 14 Crayfish*, 16(1), 47–50.
- 15 Jowett, I. G., Parkyn, S. M., & Richardson, J. (2008). Habitat characteristics of crayfish
 16 (*Paranephrops planifrons*) in New Zealand streams using generalised additive models (GAMs).

- 17 *Hydrobiologia*, 596(1), 353–365. <https://doi.org/10.1007/s10750-007-9108-z>
- 18 Jowett, I. G., & Richardson, J. (1990). Microhabitat preferences of benthic invertebrates in a
19 New Zealand river and the development of in-stream flow-habitat models for *Deleatidium*
20 spp. *New Zealand Journal of Marine and Freshwater Research*, 24(1), 19–30. <https://doi.org/10.1080/00288330.1990.9516399>
- 21
- 22 Joy, M., David, B., & Lake, M. (2013). *New zealand freshwater fish sampling protocols: Part 1*
23 — *wadeable rivers and streams*. Massey University.
- 24 Kopf, R. K., Boutier, M., Finlayson, C. M., Hodges, K., Humphries, P., King, A., Kingsford,
25 R. T., Marshall, J., McGinness, H. M., Thresher, R., & Vanderplasschen, A. (2019).
26 Biocontrol in australia: Can a carp herpesvirus (CyHV-3) deliver safe and effective ecological
27 restoration? *Biological Invasions*, 21(6), 1857–1870. [https://doi.org/10.1007/s10530-019-](https://doi.org/10.1007/s10530-019-01967-1)
28 [01967-1](https://doi.org/10.1007/s10530-019-01967-1)
- 29 Kusabs, I. A. (2023). *Lake Rotorua alum dosing monitoring programme 2023 koura and kakahi*
30 (pp. 1–8) [Bay of Plenty Regional Council]. Kusabs & Associates Ltd.
- 31 Kusabs, I. A., Quinn, J. M., & Hamilton, D. P. (2015). Effects of benthic substrate, nutrient
32 enrichment and predatory fish on freshwater crayfish (koura, *Paranephrops planifrons*)
33 population characteristics in seven Te Arawa (Rotorua) lakes, North Island, New Zealand.
34 *Marine and Freshwater Research*, 66(7), 631–643. <https://doi.org/10.1071/mf14148>
- 35 Kusabs, I. A. K., Duggan, I. C., Bruere, A. C., Scholes, P., Te Kurapa, T. W., Grant, T. M., &
1 Burdon, F. J. (2026). Invasive catfish linked to the decline of freshwater crayfish in two
2 Aotearoa New Zealand lakes. *Biological Invasions*, 28(6). [https://doi.org/10.1007/s10530-](https://doi.org/10.1007/s10530-026-03852-0)
3 [026-03852-0](https://doi.org/10.1007/s10530-026-03852-0)
- 4 Kusabs, I. A., Hicks, B. J., Quinn, J. M., & Hamilton, D. P. (2015). Sustainable management
5 of freshwater crayfish (kōura, *Paranephrops planifrons*) in Te Arawa (Rotorua) lakes, North
6 Island, New Zealand. *Fisheries Research*, 168, 35–46. [https://doi.org/10.1016/j.fishres.](https://doi.org/10.1016/j.fishres.2015.03.015)
7 [2015.03.015](https://doi.org/10.1016/j.fishres.2015.03.015)
- 8 Kusabs, I. A., & Quinn, J. M. (2009). Use of a traditional Māori harvesting method, the tau
9 kōura, for monitoring kōura (freshwater crayfish, *Paranephrops planifrons*) in Lake Rotoiti,
10 North Island, New Zealand. *New Zealand Journal of Marine and Freshwater Research*,
11 43(3), 713–722. <https://doi.org/10.1080/00288330909510036>
- 12 Kusabs, I., & Taiaroa, R. (2015). *Kōura populations in Lake Taupō and comments on the*
13 *effects of catfish*. Ngāti Tūwharetoa Māori Trust Board. [https://doi.org/10.13140/RG.2.2.](https://doi.org/10.13140/RG.2.2.12471.89761)
14 [12471.89761](https://doi.org/10.13140/RG.2.2.12471.89761)
- 15 Lee, F., Kusabs, I. A. K., Perry, G. L. W., & MacNeil, C. (2025). Identifying refugia from the
16 synergistic threats of climate change and invasive species. *Web Ecology*, 25(2), 221–239.

- 17 <https://doi.org/10.5194/we-25-221-2025>
- 18 Lennox, R. J., Kambestad, M., Berhe, S., Birnie-Gauvin, K., Cooke, S. J., Dahlmo, L. S., Eldøy,
19 S. H., Davidsen, J. G., Hanssen, E. M., Sortland, L. K., Shea, D., Nilsen, C. I., Skoglund,
20 H., Velle, G., Hellström, G., Lundberg, P., Junge, C., Ferter, K., & Wiik Vollset, K. (2025).
21 The role of habitat in predator–prey dynamics with applications to restoration. *Restoration*
22 *Ecology*, 33(3). <https://doi.org/10.1111/rec.14354>
- 23 Lenth, R. V., & Piaskowski, J. (2017). Emmeans: Estimated marginal means, aka least-squares
24 means. In *CRAN: Contributed Packages*. The R Foundation. [https://doi.org/10.32614/](https://doi.org/10.32614/cran.package.emmeans)
25 [cran.package.emmeans](https://doi.org/10.32614/cran.package.emmeans)
- 26 MacNeil, C., Lee, F., Kusabs, I., & Holmes, R. (2025). The biter bit; scavenging behaviour
27 of native freshwater crayfish on the carrion of native and introduced fish predators in
28 Aotearoa-New Zealand. *Knowledge & Management of Aquatic Ecosystems*, (426), 29.
29 <https://doi.org/10.1051/kmae/2025022>
- 30 Minns, C. K., Kelso, J. R., & Randall, R. G. (1996). Detecting the response of fish to habitat
31 alterations in freshwater ecosystems. *Canadian Journal of Fisheries and Aquatic Sciences*,
32 53(S1), 403–414. <https://doi.org/10.1139/f95-262>
- 33 Momot, W. T. (1995). Redefining the role of crayfish in aquatic ecosystems. *Reviews in*
34 *Fisheries Science*, 3(1), 33–63. <https://doi.org/10.1080/10641269509388566>
- 35 Moore, T., & McBride, C. (2026). *Aemetools: Tools to expand the Aquatic Ecosystem Model*
1 *Ensemble Package*. <https://limnotrack.com/aemetools/>
- 2 Morelli, T. L., Daly, C., Dobrowski, S. Z., Dulen, D. M., Ebersole, J. L., Jackson, S. T.,
3 Lundquist, J. D., Millar, C. I., Maher, S. P., Monahan, W. B., Nydick, K. R., Redmond, K.
4 T., Sawyer, S. C., Stock, S., & Beissinger, S. R. (2016). Managing climate change refugia
5 for climate adaptation. *PLOS ONE*, 11(8), e0159909. [https://doi.org/10.1371/journal.pone.](https://doi.org/10.1371/journal.pone.0159909)
6 [0159909](https://doi.org/10.1371/journal.pone.0159909)
- 7 Ngwenya, D. K., Geerts, S., Holmes, P. M., & Esler, K. J. (2025). Conceptualizing and
8 contextualizing “large-scale” and “scaling-up” ecological restoration. *Restoration Ecology*,
9 34(3). <https://doi.org/10.1111/rec.70266>
- 10 Nyström, P., Stenroth, P., Holmqvist, N., Berglund, O., Larsson, P., & Granéli, W. (2006).
11 Crayfish in lakes and streams: Individual and population responses to predation, productivity
12 and substratum availability. *Freshwater Biology*, 51(11), 2096–2113. [https://doi.org/10.](https://doi.org/10.1111/j.1365-2427.2006.01641.x)
13 [1111/j.1365-2427.2006.01641.x](https://doi.org/10.1111/j.1365-2427.2006.01641.x)
- 14 Orr, J. A., Macaulay, S. J., Mordente, A., Burgess, B., Albin, D., Hunn, J. G., Restrepo-Sulez,
15 K., Wilson, R., Schechner, A., Robertson, A. M., Lee, B., Stuparyk, B. R., Singh, D.,
16 O’Loughlin, I., Piggott, J. J., Zhu, J., Dinh, K. V., Archer, L. C., Penk, M., ... Jackson, M.

- C. (2024). Studying interactions among anthropogenic stressors in freshwater ecosystems: A systematic review of 2396 multiple-stressor experiments. *Ecology Letters*, 27(6). <https://doi.org/10.1111/ele.14463>
- Parkyn, S. M., & Collier, K. J. (2004). Interaction of Press and Pulse disturbance on crayfish populations: Flood impacts in pasture and forest streams. *Hydrobiologia*, 527(1), 113–124. <https://doi.org/10.1023/b:hydr.0000043189.91134.94>
- Parkyn, S. M., Collier, K. J., & Hicks, B. J. (2001). New Zealand stream crayfish: Functional omnivores but trophic predators? *Freshwater Biology*, 46(5), 641–652. <https://doi.org/10.1046/j.1365-2427.2001.00702.x>
- Parkyn, S. M., Meleason, M. A., & Davies-Colley, R. J. (2009). Wood enhances crayfish (*Paranephrops planifrons*) habitat in a forested stream. *New Zealand Journal of Marine and Freshwater Research*, 43(3), 689–700. <https://doi.org/10.1080/00288330909510034>
- Pilotto, F., Nilsson, C., Polvi, L. E., & McKie, B. G. (2018). First signs of macroinvertebrate recovery following enhanced restoration of boreal streams used for timber floating. *Ecological Applications*, 28(2), 587–597. <https://doi.org/10.1002/eap.1672>
- Prentice, M. J., Allan, M. G., Bruere, A., & Özkundakci, D. (2026). Efficacy of an inflow diversion wall for managing eutrophication in two hydrologically connected lakes: A modelling study. *Ecological Engineering*, 223, 107862. <https://doi.org/10.1016/j.ecoleng.2025.107862>
- Quinn, G. P., & Keough, M. J. (2002). *Experimental design and data analysis for biologists*. Cambridge University Press.
- R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>
- Raven, O. V., Burdon, F. J., Kusabs, I. A. K., Holmes, R., & Özkundakci, D. (2026b). Littoral habitat structure influences freshwater crayfish populations in five volcanic lakes of Aotearoa New Zealand. *Hydrobiologia*. <https://doi.org/10.1007/s10750-026-06302-z>
- Raven, O. V., MacNeil, C., Lee, F., Holmes, R., Kusabs, I. A. K., Burdon, F. J., & Özkundakci, D. (2026a). *Fear of the new: A predatory invader provokes stronger non-consumptive effects on a keystone native prey than a native predator*. <https://doi.org/10.21203/rs.3.rs-9891519/v1>
- Richman, N. I., Böhm, M., Adams, S. B., Alvarez, F., Bergey, E. A., Bunn, J. J. S., Burnham, Q., Cordeiro, J., Coughran, J., Crandall, K. A., Dawkins, K. L., DiStefano, R. J., Doran, N. E., Edsman, L., Eversole, A. G., Füreder, L., Furse, J. M., Gherardi, F., Hamr, P., ... Collen, B. (2015). Multiple drivers of decline in the global status of freshwater crayfish (Decapoda: Astacidea). *Philosophical Transactions of the Royal Society B: Biological Sciences*, 370(1662), 20140060. <https://doi.org/10.1098/rstb.2014.0060>

- 17 Ruhí, A., Herrmann, J., Gascón, S., Sala, J., Geijer, J., & Boix, D. (2012). Change in biological
18 traits and community structure of macroinvertebrates through primary succession in a man-
19 made Swedish wetland. *Freshwater Science*, 31(1), 22–37. <https://doi.org/10.1899/11-018.1>
- 20 Sayer, C. A., Fernando, E., Jimenez, R. R., Macfarlane, N. B. W., Rapacciuolo, G., Böhm, M.,
21 Brooks, T. M., Contreras-MacBeath, T., Cox, N. A., Harrison, I., Hoffmann, M., Jenkins, R.,
22 Smith, K. G., Vié, J.-C., Abbott, J. C., Allen, D. J., Allen, G. R., Barrios, V., Boudot, J.-P.,
23 ... Darwall, W. R. T. (2025). One-quarter of freshwater fauna threatened with extinction.
24 *Nature*, 638(8049), 138–145. <https://doi.org/10.1038/s41586-024-08375-z>
- 25 Selwood, K. E., & Zimmer, H. C. (2020). Refuges for biodiversity conservation: A review of
26 the evidence. *Biological Conservation*, 245, 108502. [https://doi.org/10.1016/j.biocon.2020.](https://doi.org/10.1016/j.biocon.2020.108502)
27 [108502](https://doi.org/10.1016/j.biocon.2020.108502)
- 28 Spears, B. M., Chapman, D. S., Carvalho, L., Feld, C. K., Gessner, M. O., Piggott, J. J.,
29 Banin, L. F., Gutiérrez-Cánovas, C., Solheim, A. L., Richardson, J. A., Schinegger, R.,
30 Segurado, P., Thackeray, S. J., & Birk, S. (2021). Making waves. Bridging theory and
31 practice towards multiple stressor management in freshwater ecosystems. *Water Research*,
32 196, 116981. <https://doi.org/10.1016/j.watres.2021.116981>
- 33 Streissl, F., & Hödl, W. (2002). Habitat and shelter requirements of the stone crayfish,
34 *Austropotamobius torrentium* schrank. *Hydrobiologia*, 477, 195–199. [https://doi.org/10.](https://doi.org/10.1023/A:1021094309738)
35 [1023/A:1021094309738](https://doi.org/10.1023/A:1021094309738)
- 1 Teem, J. L., Alphey, L., Descamps, S., Edgington, M. P., Edwards, O., Gemmell, N., Harvey-
2 Samuel, T., Melnick, R. L., Oh, K. P., Piaggio, A. J., Saah, J. R., Schill, D., Thomas,
3 P., Smith, T., & Roberts, A. (2020). Genetic biocontrol for invasive species. *Frontiers in*
4 *Bioengineering and Biotechnology*, 8. <https://doi.org/10.3389/fbioe.2020.00452>
- 5 Testa, S., Collier, K. J., & Death, R. G. (2011). Macroinvertebrate response to stream
6 restoration by large wood addition. *Ecohydrology*, 4(6), 800–810. [https://doi.org/10.1002/](https://doi.org/10.1002/eco.146)
7 [eco.146](https://doi.org/10.1002/eco.146)
- 8 Thresher, R. E., Hayes, K., Bax, N. J., Teem, J., Benfey, T. J., & Gould, F. (2014). Genetic
9 control of invasive fish: Technological options and its role in integrated pest management.
10 *Biological Invasions*, 16(6), 1201–1216. <https://doi.org/10.1007/s10530-013-0477-0>
- 11 Underwood, A. J. (1994). On beyond BACI: Sampling designs that might reliably detect
12 environmental disturbances. *Ecological Applications*, 4(1), 3–15. [https://doi.org/10.2307/](https://doi.org/10.2307/1942110)
13 [1942110](https://doi.org/10.2307/1942110)
- 14 Usio, N., & Townsend, C. R. (2000). Distribution of the New Zealand crayfish (*Paranephrops*
15 *zealandicus*) in relation to stream physico-chemistry, predatory fish, and invertebrate
16 prey. *New Zealand Journal of Marine and Freshwater Research*, 34(3), 557–567. <https://doi.org/10.1080/00288120008839500>

- 17 [//doi.org/10.1080/00288330.2000.9516957](https://doi.org/10.1080/00288330.2000.9516957)
- 748 Vehtari, A., Gelman, A., Simpson, D., Carpenter, B., & Bürkner, P.-C. (2021). Rank-
749 normalization, folding, and localization: An improved $\hat{R}$ for assessing convergence of
750 MCMC (with discussion). *Bayesian Analysis*, 16(2). <https://doi.org/10.1214/20-ba1221>
- 751 Verburg, P., & Albert, A. (2019). *Lake Taupo long-term monitoring programme 2017–2018*
752 (Waikato Regional Council Technical Report 2019/18; p. 179). National Institute of Water
753 & Atmospheric Research Ltd.
- 754 Vincent, W. F., Gibbs, M. M., & Dryden, S. J. (1984). Accelerated eutrophication in a New
755 Zealand lake: Lake Rotoiti, central North Island. *New Zealand Journal of Marine and*
756 *Freshwater Research*, 18(4), 431–440. <https://doi.org/10.1080/00288330.1984.9516064>
- 757 Wentworth, C. K. (1922). A scale of grade and class terms for clastic sediments. *The Journal*
758 *of Geology*, 30(5), 377–392. <https://doi.org/10.1086/622910>
- 759 Wetzel, R. G. (2001). *Limnology: Lake and river ecosystems* (3rd ed.). Academic Press.
- 760 Woolway, R. I., Sharma, S., Weyhenmeyer, G. A., Debolskiy, A., Golub, M., Mercado-Bettín,
761 D., Perroud, M., Stepanenko, V., Tan, Z., Grant, L., Ladwig, R., Mesman, J., Moore, T.
762 N., Shatwell, T., Vanderkelen, I., Austin, J. A., DeGasperi, C. L., Dokulil, M., La Fuente,
763 S., ... Jennings, E. (2021). Phenological shifts in lake stratification under climate change.
764 *Nature Communications*, 12(1). <https://doi.org/10.1038/s41467-021-22657-4>
- 765 Zhao, M., Feng, G., Wang, H., Shen, C., Fu, Y., Zhang, Y., Zhang, H., Yao, Y., Chen, J., &
766 Xu, W. (2024). The influence of shelter type and coverage on crayfish (*Procambarus clarkii*)
767 predation by catfish (*Silurus asotus*): A controlled environment study. *Animals*, 14(8), 1147.
768 <https://doi.org/10.3390/ani14081147>